PhD Notes

Version 1.0

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Contents

.

Nomenclature

Greek Symbols

λ Lagrange multiplier

μ_j Chemical potential of species j

μ_j° Standard-state chemical potential of species j

Calligraphic Symbols

$\mathcal{L}$ The Lagrangian

$\mathcal{M}_j$ Molecular weight of species j in kg / kg-mole

$\mathcal{N}$ Total kg-moles per unit mass.

$\mathcal{N}_j$ Number of kg-moles per unit mass of species j.

Roman Symbols

$\hat{R}$ Universal gas constant ($\approx$ 8314 kJ/kg-K)

ρ Density (kg/m^3)

a_{ij} Stoichiometric coefficients, e.g. number of atoms of element i in species j. Weighted by element i's molecular weight

b_i Sum of contributions from each species towards constraint for element i, i.e. $\sum_{j=1}^{\text{NS}} a_{i,j}\mathcal{N}_j$

b_i° Specified mass fraction of element i

$c_{p,j}^\circ$ Standard state specific heat of species j

$c_{v,j}^\circ$ Standard state specific heat of species j

F Helmholtz free-energy

G Gibbs free-energy

h' Sum of contributions from each species towards enthalpy constraint, i.e. $\sum_j^{\text{NS}} \mathcal{N}_j H_j^\circ$

h'_0 User-specified enthalpy (J/kg)

H_j° Standard state enthalpy of species j (J/kg)

p_{ref} Reference pressure (typically 1 atm or 1 bar).

q_j Charge of species j (0 for neutrals, 1 for ions, -1 for electrons)

T Temperature (K)

u' Sum of contributions from each species towards internal energy constraint, i.e. $\sum_j^{\text{NS}} \mathcal{N}_j U_j^\circ$

u'_0 User-specified internal energy (J/kg)

U_j° Standard state internal energy of species j (J/kg)

V Specific volume (m^3/kg)

1 Thermodynamics

1.1 Definitions

Definition 1.1 ***(Moles / molecular weight)*:** One mole of a substance contains Avogadro's number of particles. The molecular weight of a substance is the mass of one mole of that substance. In this text it is denoted as $\mathcal{M}$ and is typically given in units of kg / kg-mole or g / gm-mole. For example, the molecular weight of O_2 is 32 kg / kg-mole, and the molecular weight of N_2 is 28 kg / kg-mole.

Definition 1.2 ***(Avagadro's constant)*:** Avogradro's constant is the number of particles in one mole of that substance. It is denoted as '$\mathrm{N_A}$' and is equal to $6.02 \cdot 10^{26}$ particles per kg-mole.

Definition 1.3 ***(Universal gas constant)*:** The universal gas constant is denoted by $\hat{R}$ and is equal to 8.314 J/mol-K. Its units are 'energy per amount of substance per temperature increment' as it is a proportionality constant that relates energy to temperature and amount of substance.

Definition 1.4 ***(Boltzmann constant)*:** The Boltzmann constant is denoted by k_B and is equal to $1.381 \cdot 10^{-23}$ J/K. It is the equivalent of the universal gas constant but on a *per particle* basis.

Definition 1.5 ***(Dalton's law of partial pressures)*:** The pressure of a system can be found my summing up the partial pressures of each individual species:

$$p = \sum_j p_j \tag{1.1}$$

Definition 1.6 ***(Mass fractions)*:** The mass fraction of species j (Y_j) is defined as the ratio of the density of species j to the density of the mixture:

$$Y_j = \frac{\rho_j}{\rho} \tag{1.2}$$

Definition 1.7 ***(Molar fractions)*:** The molar fraction of species j (X_j) can be taken as the ratio of the partial pressure of species j to the pressure of the mixture, or the ratio of the number of moles of species j to the total number of moles in the system:

$$\frac{p_j}{p} = \frac{\mathcal{N}_j}{\mathcal{N}} = X_j \tag{1.3}$$

Definition 1.8 ***(Mixture moleular weight)*:** The mixture molecular weight, $\bar{\mathcal{M}}$ can be defined as:

$$\bar{\mathcal{M}} = \frac{1}{\sum_j Y_j/\mathcal{M}_j} = \sum_j X_j \mathcal{M}_j \tag{1.4}$$

Where $\mathcal{M}_j$ is the molecular weight of species j.

Definition 1.9 ***(Molar fractions to mass fractions)*****:** The mass fraction of species j can be obtained from the molar fraction by using:

$$Y_j = X_j \frac{\mathcal{M}_j}{\mathcal{M}} \tag{1.5}$$

The sum of the mass fractions

Definition 1.10 ***(Mixture gas constant)*****:** The gas constant of the mixture can be found with:

$$\bar{R} = \sum_j Y_j R_j \tag{1.6}$$

where $\bar{R}$ is the mixture specific gas constant ($\hat{R}/\mathcal{M}$) and R_j is the specific gas constant of species j.

Definition 1.11: Mixture tranlational specific heat at constant volume The mixture translational specific heat at constant volume is:

$$\bar{c}_{v,\text{tr}} = \sum_j Y_j (c_{v,j})_{\text{tr}} \tag{1.7}$$

Definition 1.12 ***(Calorically perfect gas)*****:** A calorically perfect gas is a gas where the specific heats c_p and c_v are taken to be constant. The enthalpy is then $h = c_p T$, and the internal energy is $e = c_v T$.

Definition 1.13: Thermally perfect gas A thermally perfect gas is one there c_p and c_v are variables are are functions of T only. Differential changes in enthalpy and internal energy is then $\mathrm{d}h = c_p \mathrm{d}T$, and $\mathrm{d}e = c_v \mathrm{d}T$

1.2 Equations

1.2.1 Equations of state

$$PV = mRT \tag{1.8}$$
$$pv = RT \tag{1.9}$$
$$pV = \mathcal{N}\hat{R}T \tag{1.10}$$
$$p = \rho RT \tag{1.11}$$
$$pV = \mathcal{N}\hat{R}T \tag{1.12}$$
$$p\mathcal{V} = \hat{R}T \tag{1.13}$$
$$p = C\hat{R}T \tag{1.14}$$
$$pv = \eta\mathcal{R}T \tag{1.15}$$
$$pV = NkT \tag{1.16}$$
$$p = nkT \tag{1.17}$$

Where

- P is the pressure
- V is the volume
- v is the specific gas constant
- m is the mass
- T is the temperature
- ρ is the density
- $\mathcal{V}$ is the molar volume (volume per mole)
- C is the concentration of moles per unit volume.
- η is the mole-mass ratio
- n is the number density, or number of particles per unit volume.

2 Statistical thermodynamics

Definition 2.1 ***(Bose-Einstein (BE) statistics)*:** For atoms and molecults made up of an *even* number of elementary particles such as electrons, protons, and neutrons, there is no limit to the number particles that can be in any energy state at a given instant.

Definition 2.2 ***(Fermi-Dirac (FD) statistics)*:** For atoms and molecules made up of an *odd* number of elementary particles, no energy state can be occupied by more than one particle.

Definition 2.3 ($W(N_j)$**):** For these statistics, the number of microstates in a given macrostate is given by:

$$W(N_j) = \prod_j \frac{(N_j + C_j - 1)!}{(C_j - 1)!N_j!} \tag{2.1a}$$

$$W(N_j) = \prod_j \frac{C_j!}{(C_j - N_j)!N_j!} \tag{2.1b}$$

Here, Eq. [2.1a] is used for BE statistics while Eg. [2.1b] is used for FD statistics. C_j is the number of energy states, and N_j is the number of particles in a group.

Definition 2.4 (W_{max}**):** $W_{\max}$ is defined as the largest of the individual terms in Equations [2.1] and is sometimes refered to as the *most probable microstate*. By solving for this value by setting $\delta(\ln W) = 0$, we can find the particular values of N_j that maximize W, giving:

$$N_j^* = \frac{C_j}{e^{\alpha+\beta\epsilon_j} \pm 1} \tag{2.2}$$

Where the minus sign is for BE, and the plus for FD. α and β are Lagrange multiplers used to enforce the constraints:

$$\sum_j \delta N_j = 0, \quad \sum_j \epsilon_j = \delta N_j \tag{2.3}$$

Definition 2.5 *(Boltzmann limit)*: The Boltzmann limit is used for conditions where the number of energy states far out numbers the number of group of particles. Therefore, $C_j >> N_j$, and Eqn. [2.2] reduced to:

$$N_j^* = C_j e^{-\alpha-\beta\epsilon_j} \tag{2.4}$$

Definition 2.6 *(Boltzmann distribution)*: The Boltzmann distribution, after solving for α and β in the above, gives:

$$N_j^* = N\frac{C_j e^{-\epsilon_j/kT}}{\sum_j C_j e^{-\epsilon_j/kT}} \tag{2.5}$$

Definition 2.7 *(Molecular partition function)*: The molecular partition function, Q, is defined as the denominator in Eq. [2.5]:

$$Q = \sum_j C_j e^{-\epsilon_j/kT} \tag{2.6}$$

However, this is in terms of groups of energy states having the same energy level ϵ_j. If we we transition to the use of individual energy states, ϵ_j becomes ϵ_i, and the C_j term is neglected. The equation becomes:

$$Q = \sum_i e^{-\epsilon_i/kT} \tag{2.7}$$

Definition 2.8 *(Population of energy states)*: The population of energy state ϵ_i then becomes:

$$N_i^{*'} = N\frac{e^{-\epsilon_i/kT}}{\sum_i e^{-\epsilon_i/kT}} \tag{2.8}$$

Definition 2.9 *(Energy levels and degeneracy)*: Energy levels are sometimes more convenient to use. An energy level can have multiple energy states in it. If a level contains more than one state, it is said to be *degnerate*, and the number of states inside an energy level ϵ_l is given as g_l. Therefore, the population of particles in energy level ϵ_l, and the partition function is given as:

$$Q = \sum_l g_l e^{-\epsilon_l/kT} \tag{2.9}$$

$$N_l^{*'} = N \frac{g_l e^{-\epsilon_l/kT}}{\sum_i g_l e^{-\epsilon_i/kT}} \tag{2.10}$$

Definition 2.10 *(Boltzmann's relation)*: Boltzmann's relation finds the relationship between the number of microstates Ω and the entropy of a system as:

$$S = k \ln \Omega \tag{2.11}$$

Definition 2.11 *(Entropy)*: Entropy can then be shown to be of the form:

$$S = Nk \left(\ln \frac{Q}{N} + 1 \right) + \frac{E}{T} \tag{2.12}$$

Definition 2.12 *(Helmholtz free energy)*: The Helmholtz free energy, F, is defined as:

$$F = E - TS \tag{2.13}$$

Here, E is the internal energy, T is the temperature, and S the entropy of the system. Differentiation of [2.13] yields:

$$dF = dE - TdS - SdT \tag{2.14}$$

Since $dE - TdS = \mu dN - pdV$, (μ is the chemical potental per molecule, N is the number of particles, and p, V is the pressure and volume of the system), this equation can be rewritten as:

$$dF = -SdT - pdV + \mu dN \tag{2.15}$$

Which yields the following results:

$$S = -\left(\frac{\partial F}{\partial T}\right)_{V,N} \qquad p = -\left(\frac{\partial F}{\partial V}\right)_{T,N} \qquad \mu = \left(\frac{\partial F}{\partial N}\right)_{T,V} \tag{2.16}$$

Definition 2.13 *(Thermodynamic properties of a system)*: The thermodynamic properties of a system, assuming Q and ϵ_i's are independant of N, are:

$$F = -NkT\left(\ln\frac{Q}{N} + 1\right) \tag{2.17a}$$

$$S = Nk\left[\ln\frac{Q}{N} + 1 + T\frac{\partial(\ln Q)}{\partial T}\right] \tag{2.17b}$$

$$E = NkT^2\frac{\partial(\ln Q)}{\partial T} \tag{2.17c}$$

$$p = NkT\frac{\partial(\ln Q)}{\partial V} \tag{2.17d}$$

$$\mu = -kT\ln\frac{Q}{T} \tag{2.17e}$$

Definition 2.14 ***(Total partion function and energy)*****:** As will be defined shortly, the total partion function is the product of all energy-type specific partion functions, i.e.:

$$Q = Q_{tr}Q_{rot}Q_{vib}Q_{el} = Q_{tr}\cdot\prod_{int}Q_{int} \tag{2.18}$$

The total internal energy of a molecule is:

$$\epsilon = \epsilon_{tr} + \epsilon_{rot} + \epsilon_{vib} + \epsilon_{el} \tag{2.19}$$

This also works for specific heat, c_v.

Definition 2.15 ***(Partition function for translation energy)*****:** The partition function for translation energy is:

$$Q_{tr} = V\left(\frac{2\pi mkT}{h^2}\right)^{\frac{3}{2}} \tag{2.20}$$

The translational internal energy can be found as:

$$E_{tr} = \frac{3}{2}NkT \qquad e_{tr} = \frac{3}{2}RT \tag{2.21}$$

Where the first equation is per N molecules, and the second equation is on a per-unit mass basis. The specific heat is then:

$$c_{v_{tr}} = \left(\frac{\partial e_{tr}}{\partial T}\right)_v = \frac{3}{2}R \tag{2.22}$$

2.1 Electronic energies

Since monoatomic molecules do not have rotational or vibrational energies, the only internal energy is the electronic one. However, this still applies to diatomic molecules.

Definition 2.16 ***(Electronic internal energy)*****:** The electronic internal energy can largely be ignored at moderate temperatures for some species of gases. However, they can become important at higher temperatures. Because the spacing between the levels can be so large, usually only a few degenerate levels are accounted for. Electronic energy accounts for when electrons populate higher electronic energy levels.

Definition 2.17 *(Electronic partition function)*: The electronic partition function is:

$$Q_{el} = \sum_l g_l e^{-\epsilon_l/kT} = \sum_l g_l e^{\theta_l/T} \tag{2.23}$$

Where $\theta_l = \epsilon_l/k$ are the *characteristic temperatures for electronic excitation*. Energy of the ground level is taken to be 0, so the first term is just g_0 with no exponential. If we assume that just the first two terms are used, the internal energy becomes:

$$e_{el} = R\theta_1 \frac{(g_1/g_0)e^{-\theta_l/T}}{1 + (g_1/g_0)e^{-\theta_1/T}} \tag{2.24}$$

and the specific heat:

$$c_{v_{el}} = R\left(\frac{\theta_1}{T}\right)^2 \frac{(g_1/g_0)e^{-\theta_1/T}}{[1 + (g_1/g_0)e^{-\theta_1/T}]^2} \tag{2.25}$$

2.2 Diatomic internal energies

3 High Temperature Transport Models

References for this section

1. *A Review of Reaction Rates and Thermodynamic and Transport Properties for an 11-Species Air Model for Chemical and Thermal Nonequilibrium Calculations to 30,000 K*, Gupta, Yos, Thompson, Lee, 1990 NASA RP 1232.
2. *An assessment of Transport Property Methodologies for Hypersonic Flows*, Palmer, 1971, AIAA.

3.1 Collision integral definitions

Definition 3.1 *(Collision cross sections)*: The collision cross section has infinitely many values. Four of them are used for fluid dynamics to determine transport properties. The collision cross sections are, in general, the weighted averages of the cross sections for collisions between species i and j (sometimes denoted s and r). They are defined as:

$$\pi\bar{\Omega}_{ij}^{(l,s)} = \frac{\int_0^\infty \int_0^\pi \exp(-\gamma^2)\gamma^{2s+3}(1-\cos^l\chi)4\pi\sigma_{ij}\sin\chi d\chi d\gamma}{\int_0^\infty \int_0^\pi \exp(-\gamma^2)\gamma^{2s+3}(1-\cos^l\chi)\sin\chi d\chi d\gamma} \tag{3.1}$$

Variables are:

- $\sigma_{ij} = \sigma_{ij}(\chi, g)$ is the differential-scattering cross section for the pair (i, j).
- χ is the scattering angle in the center of mass system.
- g is the relative velocity of the colliding particles.
- γ is the reduced velocity equal to $\left(\left[\frac{m_i m_j}{2(m_i+m_j)kT}\right]^{\frac{1}{2}}\right)g$

In the collision integral symbol, l controls how strong the contribution of the differential-scattering angle (χ) is. s is a weighting on the reduced velocity (γ) of the particles. Of course, the subscript (ii) refers to the collision integral for similar particles. Here is a list of where certain collision integrals show up:

- Viscosity (μ) is a function of $\pi\bar{\Omega}_{ij}^{(2,2)}$ only.
- Translational component of thermal conductivity (κ_{tr}) is a function of $\pi\bar{\Omega}_{ij}^{(2,2)}$ only.
- The internal component of thermal conductivity (κ_{int}) is a function of $\pi\bar{\Omega}_{ij}^{(1,1)}$ only or a function of $\pi\bar{\Omega}_{ij}^{(1,1)}$ and $\pi\bar{\Omega}_{ij}^{(2,2)}$, depending on the expression used.
- The diffusion coefficient D_{ij} is a function of $\pi\bar{\Omega}_{ij}^{(1,1)}$ only.

Definition 3.2 *(Gupta's curve fits for collision integrals)*: Gupta utilizes curve fits for the collision integrals. They have the following form:

$$\pi\bar{\Omega}_{ij}^{(l,s)} = [\exp D]\, T^{\left[A(\ln T)^2 + B\ln T + C\right]} \tag{3.2}$$

$$B_{i,j}^{*(2,2)} = [\exp C]\, T^{\left[A(\ln T)^2 + B \ln T\right]} \tag{3.3}$$

Definition 3.3 ***(Chapman-Enskog procedure)*:**

3.2 Transport properties of a single species

Definition 3.4 ***(Viscosity of a single species)*:** The viscosity of a gas containing a single species is:

$$\mu_i = \frac{5}{16} \frac{\sqrt{\pi m_i k T}}{\pi \bar{\Omega}_{ii}^{(2,2)}} = 8.3861 \cdot 10^{-6} \frac{\sqrt{\mathcal{M}_i T}}{\pi \bar{\Omega}_{ij}^{(2,2)}} \qquad \text{units} = \frac{\text{N - s}}{\text{m}^2} \tag{3.4}$$

Where $\mathcal{M}_i$ is the molecular weight of species i, and T is the temperature in Kelvin.

Definition 3.5 ***(Thermal conductivity of a single species)*:** For the thermal conductivity of a single species, we have a number of contributions. The total κ of the species is the sum of its internal energy specific contributions, i.e.:

$$\kappa = \kappa_{\text{tr}} + \kappa_{\text{rot}} + \kappa_{\text{vib}} + \kappa_{\text{el}} = \kappa_{\text{tr}} + \kappa_{\text{int}} \tag{3.5}$$

Where the internal energies that are not translational have been grouped into κ_{int}.

$$\kappa_{\text{tr},i} = \frac{5}{2} \frac{\mu_i}{\mathcal{M}_i} (c_{v,i})_{\text{tr}} = \frac{15}{4} \frac{\hat{R}}{\mathcal{M}_i} \mu_i \qquad \text{units} = \frac{\text{J}}{\text{m-s-K}} \tag{3.6}$$

$$\kappa_{\text{int},i} = \frac{6}{5} \frac{c_{p_{\text{int},i}}}{\hat{R}} \frac{\hat{R}}{\mathcal{M}_i} \mu_i \frac{\pi \bar{\Omega}_{ii}^{(2,2)}}{\pi \bar{\Omega}_{ii}^{(1,1)}} \tag{3.7}$$

Here, $c_{p_{\text{int},i}}$ is the total specific heat with the translation specific heat subtracted.

3.3 Transport properties of a mixture

For species summation and Wilke's mixing rule, mu_j and κ_j are calculated using the above equations for single species viscosity and conductivity.

Definition 3.6 ***(Species summation)*:** The concept of the species summation version of finding mixture transport properties is quite simple. It assumes no interaction between species and is expressed as:

$$\mu = \sum_{j=1}^{\text{NS}} X_j \mu_j \qquad \kappa = \sum_{j=1}^{\text{NS}} X_j \kappa_j \tag{3.8}$$

Definition 3.7 ***(Wilke's mixing rule)*:** Both viscosity and thermal conductivity can be approximated using Wilke's mixing rule. It uses a quantity, ϕ_i, which is expressed as:

$$\phi_i = \sum_{j=1}^{\mathrm{NS}} \frac{X_j \left[1 + \sqrt{\frac{\mu_i}{\mu_j}} \left(\frac{\mathcal{M}_j}{\mathcal{M}_i}\right)^{\frac{1}{4}}\right]^2}{\sqrt{8\left(1 + \frac{\mathcal{M}_i}{\mathcal{M}_j}\right)}} = \sum_{j=1}^{\mathrm{NS}} \frac{X_j \left[1 + \sqrt{\frac{\pi\bar{\Omega}_{jj}^{(2,2)}}{\pi\bar{\Omega}_{ii}^{(2,2)}}}\right]^2}{\sqrt{8\left(1 + \frac{\mathcal{M}_i}{\mathcal{M}_j}\right)}} \tag{3.9}$$

Then the viscosity and thermal conductivity is found as:

$$\mu = \sum_{j=1}^{\mathrm{NS}} \frac{X_j \mu_j}{\phi_j} \qquad \kappa = \sum_{j=1}^{\mathrm{NS}} \frac{X_j \kappa_j}{\phi_j} \tag{3.10}$$

Definition 3.8 ***(Chapman-Enskog approximation method)*****:** For obtaining mixture transport quantities using Chapman-Enskog approximations, a number of new variables are used. This is the most rigorous method for finding the properties. The units of Δ is in centimeters-second, while α_{ij} is unitless.

$$\Delta_{ij}^{(1)} = \frac{8}{3}(1.5460 \cdot 10^{-20}) \left[\frac{2\mathcal{M}_i \mathcal{M}_j}{\pi N_A kT(\mathcal{M}_i + \mathcal{M}_j)}\right]^{\frac{1}{2}} \pi\bar{\Omega}_{ij}^{(1,1)} \tag{3.11}$$

$$\Delta_{ij}^{(2)} = \frac{16}{5}(1.5460 \cdot 10^{-20}) \left[\frac{2\mathcal{M}_i \mathcal{M}_j}{\pi N_A kT(\mathcal{M}_i + \mathcal{M}_j)}\right]^{\frac{1}{2}} \pi\bar{\Omega}_{ij}^{(2,2)} \tag{3.12}$$

For electron-electron collisions, we have:

$$\pi\bar{\Omega}_{ee}^{(2,2)} = 1.29 Q_e \cdot 10^{16} \tag{3.13}$$

For ion-ion collisions, we have:

$$\pi\bar{\Omega}_{II}^{(2,2)} = 1.36 Z^4 Q_e \cdot 10^{16} \tag{3.14}$$

Then for both:

$$\pi\bar{\Omega}_{ee}^{(1,1)} = \pi\bar{\Omega}_{II}^{(1,1)} = 0.795 Z^4 Q_e \cdot 10^{16} \tag{3.15}$$

Here, $Z = 1$ for singly ionized species (for aerospace applications this is fine), and:

$$Q_e = \frac{e^4}{(kT)^2} \ln\Lambda \qquad \mathrm{cm}^2 \tag{3.16}$$

$$\Lambda = \left[\frac{9(kT)^3}{4\pi e^6 n_e} + \frac{16(kT)^2}{e^4 n_e^{2/3}}\right]^{1/2} \tag{3.17}$$

$$= \left[2.09 \cdot 10^{-2}\left(\frac{T^4}{10^{12} p_e}\right) + 1.52\left(\frac{T^4}{10^{12} p_e}\right)^{2/3}\right]^{1/2} \tag{3.18}$$

Here, p_e is the electron pressure in atmospheres, n_e is the number density of electrons, and $e = 4.8 \cdot 10^{-10}$ esu. These equations are only useful unter for $p_e <$ some limiting value, which should be true in aerospace applications:

$$p_{em} = 0.0975 \left(\frac{T}{10^3} \right)^4 \tag{3.19}$$

Depending on the data given, the Coulomb cross section may need to be altered. Gupta and Yos give collision cross sections at the above limiting electron pressure, so at any other pressure the cross section needs to be changed:

$$\frac{\pi\bar{\Omega}_{ij}^{(l,s)}(p_e)}{\pi\bar{\Omega}_{ij}^{(l,s)}(p_{em})} = \ln \Lambda(p_e) \tag{3.20}$$

$$= \frac{1}{2} \ln \left[2.09 \cdot 10^{-2} \left(\frac{T}{1000 p_e^{1/4}} \right)^4 + 1.52 \left(\frac{T}{1000 p_e^{1/4}} \right)^{8/3} \right] \tag{3.21}$$

Finally, α_{ij} is needed for the heavy-particle translational component of thermal conductivity:

$$\alpha_{ij} = 1 + \frac{[1 - (\mathcal{M}_i/\mathcal{M}_j)][0.45 - 2.54(\mathcal{M}_i/\mathcal{M}_j)]}{[1 + (\mathcal{M}_i/\mathcal{M}_j)]^2} \tag{3.22}$$

Assuming thermochemical nonequilibrium, the mixture viscosity *appears* to not be a function of vibrational or rotation energy, and is thus found with:

$$\mu = \sum_{i=1}^{\text{NS}-1} \left[\frac{\frac{\mathcal{M}_i}{N_A} X_i}{\sum_{j=1}^{\text{NS - 1}} X_j \Delta_{ij}^{(2)}(T) + X_e \Delta_{ie}^{(2)}(T_e)} \right] + \frac{\frac{\mathcal{M}_e}{N_A} X_e}{\sum_{j=1}^{\text{NS}} X_j \Delta_{ej}^{(2)}(T_e)} \tag{3.23}$$

Where X_j is the molar fraction, T is the Temperature, and T_e is the electron temperature. In constrast to viscosity, the thermal conductivity is dependant on all energy modes, and therefore has contributions from each mode. They are:

Translational component without electron-heavy-particle and electron-electron collision:

$$K_{\text{tr}}^* = (2.3901 \cdot 10^{-8}) \frac{15}{4} k \sum_{i=1}^{\text{NS-1}} \left[\frac{X_i}{\sum_{j=1} \alpha_{ij} \Delta_{ij}^{(2)}(T) + 3.54 X_e \Delta_{ie}^{(2)}(T_e)} \right] \tag{3.24}$$

Rotational component. "p.e." means "partial excitation" while "f.e." mean "full excitation". If it is in equilibrium with translational temperature, $T_{\text{rot}} = T$:

$$(K_{\text{rot}})_{\text{p.e.}} = (2.3901 \cdot 10^{-8}) k \sum_{\text{i=mol.}} \left[\frac{\left[\frac{c_{p,i}(T_{\text{rot}})}{\hat{R}} - \frac{5}{2} \right] X_i}{\sum_{j=1}^{\text{NS-1}} X_j \Delta_{ij}^{(1)}(T) + X_e \Delta_{ie}^{(1)}(T_e)} \right] \tag{3.25}$$

$$(K_{\text{rot}})_{\text{f.e.}} = (2.3901 \cdot 10^{-8}) k \sum_{\text{i=mol.}} \left[\frac{X_i}{\sum_{j=1}^{\text{NS-1}} X_j \Delta_{ij}^{(1)}(T) + X_e \Delta_{ie}^{(1)}(T_e)} \right] \tag{3.26}$$

Vibrational component:

$$(K_{\text{vib}})_{\text{p.e.}} = (2.3901 \cdot 10^{-8})k \sum_{\text{i=mol.}} \left[\frac{\left[\frac{c_{p,i}(T_{\text{vib}})}{\hat{R}} - \frac{7}{2}\right] X_i}{\sum_{j=1}^{\text{NS-1}} X_j \Delta_{ij}^{(1)}(T) + X_e \Delta_{ie}^{(1)}(T_e)} \right] \tag{3.27}$$

$$(K_{\text{vib}})_{\text{f.e.}} = (2.3901 \cdot 10^{-8})k \sum_{\text{i=mol.}} \left[\frac{X_i}{\sum_{j=1}^{\text{NS-1}} X_j \Delta_{ij}^{(1)}(T) + X_e \Delta_{ie}^{(1)}(T_e)} \right] \tag{3.28}$$

Electronic component:

$$K_{\text{el}} = (2.3901 \cdot 10^{-8})k \sum_{\text{i=1}}^{\text{NS-1}} \left[\frac{\left[\frac{c_{p,i}(T_{\text{el}})}{\hat{R}} - \frac{9}{2}\right] X_i}{\sum_{j=1}^{\text{NS-1}} X_j \Delta_{ij}^{(1)}(T) + X_e \Delta_{ie}^{(1)}(T_e)} \right] \tag{3.29}$$

Free electron component:

$$K_{\text{e}} = (2.3901 \cdot 10^{-8})k \frac{X_e}{\sum_{j=1}^{\text{NS-1}} X_j \Delta_{ej}^{(2)}(T_e) + X_e \Delta_{ee}^{(2)}(T_e)} \tag{3.30}$$

4 Chemical Rate Kinetics

4.1 Rate equations

Definition 4.1 *(Reactions)*: For multicomponent gas with NS reacting species and NR different reactions, the stoichiometric relations for the change in reactants and products are:

$$\sum_{i=1}^{\mathrm{NS}} \nu_i' X_i \leftrightarrows \sum_{i=1}^{\mathrm{NS}} \nu_i'' X_i, \qquad r = 1, \ldots, NR \tag{4.1}$$

ν_i' and ν_i'' are the stoichiometric coefficients for reactants and products respectively. $k_{f,r}$ governs the forward reaction (right arrow), and $k_{b,r}$ governs the backward reaction (left arrow).

Definition 4.2 *(Mass production)*: The net rate of MASS production of a species per unit volume has a number of different expressions. Consider the reaction:

$$O_2 + M \longrightarrow 2O + M \tag{4.2}$$

Here, M is an arbitrary collision partner, deemed a *third-body*. Concentration of species j is the moles of j per unit volume. $[O_2]$ is the concentration of O_2. The time rate of change of the concentration of O is then:

$$\frac{d[O]}{dt} = 2k[O_2][M] \tag{4.3}$$

Where k is the reaction rate. This equation gives the rate at which Eq. [4.2] goes from left to right. Therefore, k is actually the *forward* reaction rate, k_f. The two previous equations should then be:

$$O_2 + M \xrightarrow{k_f} 2O + M \tag{4.4}$$

$$\frac{d[O]}{dt} = 2k_f[O_2][M] \tag{4.5}$$

Of course, the right to left reaction is the *backwards* reaction and is governed by the backwards rate coefficient:

$$O_2 + M \xleftarrow[k_b]{} 2O + M \tag{4.6}$$

$$\frac{d[O]}{dt} = -2k_b[O]^2[M] \tag{4.7}$$

You must combine these forward and backwards equations because both reactions happen. This gives:

$$O_2 + M \xleftrightarrow[k_b]{k_f} 2O + M \tag{4.8}$$

$$\frac{d[O]}{dt} = 2\left(k_f[O_2][M] - k_b[O]^2[M]\right) \tag{4.9}$$

The net rate equation for an arbitrary reaction is then:

$$\frac{d[X_i]}{dt} = (\nu_i'' - \nu_i')\left\{k_f \prod_i [X_i]^{\nu_i'} - k_b \prod_i [X_i]^{\nu_i''}\right\} \tag{4.10}$$

Where ν' is the stoichiometric for a reactant species, and ν'' is for product species.

Definition 4.3 ***(Reaction rate relations)*****:** The equilibrium constant is very useful to find one reaction rate from the other. There are two equilibrium constants: K_c is the equilibrium constant based on concentrations, and K_p is the equilibrium constant based on partial pressures. They are related through:

$$K_{c,r} = \left(\frac{p_\circ}{\hat{R}T}\right)^{\nu_m} \exp\left[-\sum_s (\nu_s'' - \nu_s')\left(\frac{H_s^\circ}{\hat{R}T} - \frac{S_s^\circ}{\hat{R}}\right)\right] \tag{4.11}$$

Where it is raised to a sum due to the complicated nature of the equilibrium constant. Typically, the rate coefficient has variable units, since it depends on the number of molecules in the reaction, which gives rise to a key concept:

For most chemical kinetic models, the reaction's reactants always consist of 2 species, while the reaction's products may contain two or three species. Therefore, the forwards rate coefficients k_f should always have the same units. Since the equilibrium constant is used to find k_b, no unit conversions are necessary besides perhaps from cgs to mks units.

With the equilibrium constant of a reaction known, you can find the forward/backward reaction rates as stated above with:

$$\frac{k_{f,r}(T)}{k_{b,r}(T)} = K_{\text{c,r}} \tag{4.12}$$

Definition 4.4 ***(Arrhenius equations)*****:** Both forward and backward rate coefficients can be found using the Arrhenius or modified Arrhenius equations, though usually the forward rate is computed and the backward rate is found with the equilibrium constant. The Arrhenius equation has the form:

$$k_{f,r} = Ce^{-E_a/kT} \tag{4.13}$$

Where C is a constant, E_a is the activation energy, k is the Boltzmann constant, and T is the temperature. The modified Arrhenius form (more widely used) is:

$$k_{f,r} = CT^\eta e^{-E_0/kT} \tag{4.14}$$

Here, η is a reaction specific exponent, and E_0 depends on the reaction (could be a dissociation energy, could be 0 for recombination, etc.)

4.2 Reaction Types

Definition 4.5 *(Dissociation Reaction)*: A dissociation reaction occurs when a molucule collides with some other species and one or more atoms ”detatch” from the molecule. Examples are:

$$O_2 + M \leftrightarrows 2O + M \tag{4.15}$$
$$N_2 + M \leftrightarrows 2N + M \tag{4.16}$$
$$NO + M \leftrightarrows N + O + M \tag{4.17}$$
$$O_2 + N \leftrightarrows NO + O \tag{4.18}$$
$$N_2 + O \leftrightarrows NO + N \tag{4.19}$$
$$N_2 + O_2 \leftrightarrows 2NO \tag{4.20}$$
$$N + O \leftrightarrows NO^+ + e^- \tag{4.21}$$

Eq. [4.18] is an example of a *bimolecular exchange* reaction. Eq. [4.21] is called a *dissociative-recombination* reaction since the recombination of the NO^+ ion with an electron produces a dissociated product.

To be continued...

Definition 4.6 *(Efficiencies)*: For certain reactions like dissociation, the outcome of the reaction is the same regardless of the other reactant. These are called *third-body reactions*, since the secondary reactant does not change chemically. The third-body is usually represented with an M as we have already discussed:

$$N_2 + M \rightleftarrows N + N + M \tag{4.22}$$

This M can be any species, and the Arrhenius equation is nearly identical for any M, with only a slight change in the leading ceofficient C. This coefficient change is called an *efficiency* (denoted here as ϵ), and the algebra will be shown as to how this is beneficial to use. A list of efficiencies in certain reactions are shown in Table 1.

Table 1: Change in modified Ahhrenius constant C for some third-bodies in N_2 dissociation.

Third body	C	ϵ
$M = N_2$	7.0e21	1.0
$M = O_2$	7.0e21	1.0
$M = NO$	7.0e21	1.0
$M = N$	3.0e22	4.2857
$M = O$	3.0e22	4.2857

Since $7.0e21$ is the smallest value, this is given an efficiency of 1.0, and then any other reaction's efficiency is given as some multiplier of this base value. The physical meaning of efficiencies is how ”efficient” that third body is at making that reaction happen. ϵ is larger for atoms, since they are more efficient in sending their energy into dissociation of a molecule. We can now do some algebra on the Arrhenius equation by letting:

$$C = \epsilon C_c \tag{4.23}$$

$$\gamma = C_c T^{\eta} e^{E_0/kT} \tag{4.24}$$

The forward and backwards reaction constants are then:

$$k_f = \epsilon\gamma, \qquad k_b = \frac{\epsilon\gamma}{K_{eq}} \tag{4.25}$$

For a third body reaction, the change in the concentration of N_2 due to a single third-body M is then:

$$\begin{aligned}\frac{d[N_2]}{dt} &= (\nu'' - \nu')\left\{\epsilon\gamma[N_2][M] - \frac{\epsilon\gamma}{K_{eq}}[N]^2[M]\right\}\\ &= (\nu'' - \nu')\cdot\epsilon\cdot[M]\underbrace{\left\{\gamma[N_2] - \frac{\gamma}{K_{eq}}[N]^2\right\}}_{R}\end{aligned}$$

R is independant of the third-body, so it is a constant and only needs to be computed once. Summing over contributions from all independant reactions gives:

$$\frac{d[N_2]}{dt} = \underbrace{(0-1)}_{\nu''_{N_2}-\nu'_{N_2}}\left\{\sum_i \epsilon_i\cdot[M_i]\right\}R$$

Where:

$$\sum_i \epsilon_i[M_i] = \epsilon_{N_2}[N_2] + \epsilon_{O_2}[O_2] + \epsilon_{NO}[NO] + \epsilon_N[N] + \epsilon_O[O] + \dots \tag{4.26}$$

R is retained for the production of N atoms as well, you only need to change the coefficient from the $\nu''_i - \nu''_i$ term:

$$\frac{d[N]}{dt} = \underbrace{(2-0)}_{\nu''_N-\nu'_N}\left\{\sum_i \epsilon_i\cdot[M_i]\right\}R$$

4.3 Example rate equation solution

Here, and example is done to find the rate of production of NO. Consider the following reactions that all contain NO:

$$NO + O_2 \leftrightarrows N + O + O_2 \tag{4.27a}$$
$$NO + N_2 \leftrightarrows N + O + N_2 \tag{4.27b}$$
$$NO + NO \leftrightarrows N + O + NO \tag{4.27c}$$
$$NO + O \leftrightarrows N + O + O \tag{4.27d}$$
$$NO + N \leftrightarrows N + O + N \tag{4.27e}$$
$$O_2 + N \leftrightarrows NO + O \tag{4.27f}$$
$$N_2 + O \leftrightarrows NO + N \tag{4.27g}$$

The forward and backward rate coefficients for the reactions are $k_{f,a}$ and $k_{b,a}$ for the first reaction, all the way down with consistent lettering with the equation number. The entire equation is then:

$$\begin{aligned}\frac{d[NO]}{dt} = -&\left\{\gamma[NO] - \frac{\gamma}{K_{eq}}[N][O]\right\}\sum_i \epsilon_i[M] \\ &+ \{k_{f,f}[O_2][N] - k_{b,f}[NO][O]\} \\ &+ \{k_{f,g}[N_2][O] - k_{b,g}[NO][N]\}\end{aligned}$$

5 Internal energy relaxation

5.1 Translational-Vibrational relaxation

The source term for vibrational energy due to transfer of energy between translational and vibrational energy modes from the Landau-Teller model is:

$$Q^{\mathrm{LT}}_{t-v} = \sum_s Q^{\mathrm{LT}}_{t-v,s} \tag{5.1}$$

Where

$$Q^{\mathrm{LT}}_{t-v,s} = \rho_s \frac{e^*_{vs} - e_{vs}}{< \tau_s >} \tag{5.2}$$

e^*_{vs} is the equilibrium vibrational energy, which is evaluated at the tran-rotational tempertature, T_{tr}, while e_{vs} is the actual vibrational energy evaluated with T_v. The molar-averaged relaxation time, $< \tau_s >$ is defined by:

$$< \tau_s >= \frac{\sum_r X_r}{\sum_r X_r/\tau_{sr}} \tag{5.3}$$

This sum happens for all particles r, and molecules s. The term $\tau_s r$ is the relaxation time between species and has two parts - the Millikan-White part, as well as the high-temperature Park correction. The Millikan-White part uses the equations:

$$\tau^{\mathrm{MW}}_{sr} = \frac{101325}{p} \exp\left[A_{sr}(T^{-1/3} - B_{sr}) - 18.42\right] \tag{5.4}$$

Where $A_s r$ and B_{sr} are either set constants or computed from:

$$\mu_{sr} = \frac{\mathcal{M}_s \mathcal{M}_r}{\mathcal{M}_s + \mathcal{M}_r} \tag{5.5}$$

$$A_{sr} = 1.16 \times 10^{-3} \mu_{sr}^{1/2} \theta_{vs}^{4/3} \tag{5.6}$$

$$B_{sr} = 0.015 \mu_{sr}^{1/4} \tag{5.7}$$

The Park correction is:

$$\tau^{\mathrm{P}}_s = \frac{1}{\bar{c}_s \sigma_s n} \tag{5.8}$$

$$\bar{c}_s = \sqrt{\frac{8\hat{R}T_{tr}}{\mathcal{M}_s \pi}} \tag{5.9}$$

$$\sigma_s = \sigma_{v,s} \left(\frac{50000}{T_{tr}}\right)^2 \tag{5.10}$$

$$n = \frac{p}{k_b T_{tr}} \tag{5.11}$$

5.2 Chemical-Vibrational relaxation

$$Q_{c-v,s} = \dot{w}_s e_{v,s} \tag{5.12}$$

6 Chemical Equilibrium CFD

6.1 Minimization Procedures

6.1.1 Constraints for Both Procedures

There are two constraints that are used the most in both energy minimization procedures. The first, Eq. [6.1], is always used, while the charge constraint in Eq. [6.2] is only used when ions are present:

$$\sum_{j=1}^{\mathrm{NS}} a_{ij}\mathcal{N}_j - b_i^\circ = 0 \tag{6.1}$$

$$\sum_{j=1}^{\mathrm{NS}} q_j\mathcal{N}_j = 0 \tag{6.2}$$

Since these are used in abundance for both procedures, their constraints are placed in the Lagrangian term. Other constraints (such as specified volume, temperature, internal energy, and entropy) are given their own residual equations in the Newton solve without addition into the Lagrangian.

Eq. [6.1] says that the number of moles of element i must remain constant during the minimization process, as atoms can not be created or destroyed without nuclear reactions being involved. b_i° is the mass fraction of element i as chosen by the user. a_{ij} is the stoichiometric coefficient, i.e. how many atoms of element i are in species j. In this code, the stochiometric coefficient is actually multiplied by the element i's molecular weight.

Eq. [6.2] says that the charge of the system must remain neutral, i.e., if cations are formed, than there must be an equal number of free electrons in the gas to balance the charge.

6.1.2 The Lagrangian

The Lagrangian is defined as;

$$\mathcal{L} = f + \sum_j \lambda_j g_j \tag{6.3}$$

where f is the function we want to minimize, λ_j are Lagrange multipliers, and g_j are the constraint functions equal to 0. Instead of minimizing just f, we aim to minimize the Lagrangian $\mathcal{L}$.

6.1.3 Newton-Raphson Iterative Method

A Newton-Raphson iterative procedure is used to minimize the energies. Since this is essentially a root finding system of equations, and we are seeking the solution update, the method looks like the following after using a Taylor expansion on the function f:

$$\sum_{i=1}^{N} \frac{\partial f}{\partial x_i}\Delta x_i = -f(x) \tag{6.4}$$

Here, $f(x)$ is the function that is being minimized. In the case of Gibbs and Helmholtz energy minimization, f takes the form of $\partial\mathcal{L}/\partial x$, with x being non-linear variables used to guide the convergence of the system. The non-linear correction variables are in the form of $\Delta\ln\mathcal{N}_j$, $\Delta\ln\mathcal{N}$, $\Delta\ln T$, and $\pi_{i,q} = \Delta\pi_{i,q}$. For the last correction variable, it is argued that setting π to 0 at the beginning of each update does not influence the solution for the non-reduced equations, so the Δ is dropped for that term. This will influence the form of the update equations, which will be talked about in due time.

6.1.4 Gibbs Minimization

The Gibbs energy is a function of temperature, pressure, and composition of a gas:

$$G = G(T, p, \mathcal{N}_1, \mathcal{N}_2, ..., \mathcal{N}_{\text{NS}}) = \sum_{j=1}^{\text{NS}} \mathcal{N}_j \mu_j \tag{6.5}$$

Where NS is the number of gas species, $\mathcal{N}_j$ is the number of kg-moles of species j per kilgoram (i.e., the mole-mass fraction), and μ_j is the chemical potential of species j defined by:

$$\mu_j = \frac{\partial G}{\partial \mathcal{N}_j} = \mu_j^\circ + \hat{R}T\left[\ln\left(\frac{p}{p_{\text{ref}}}\right) + \ln\left(\frac{\mathcal{N}_j}{\mathcal{N}}\right)\right] \tag{6.6}$$

Here, $\hat{R}$ is the universal gas constant, T and p are the temperature and pressure of the system, p_{ref} is the reference pressure usually takes as 101,325 Pa, or 1 bar. This value depends on the thermodynamic data you are reading in. For NASA-9 coefficient polynomial, 1 bar is used. depending on the literature.$\mathcal{N}$ is the number of kg-moles per unit mass of the mixture. In order to find chemical equilibrium, the derivative of G wrt $\mathcal{N}_j$ is set to 0 for all species. However, a number of constraints need to be added in.

6.1.5 Constraints

Firstly, the elemental constraint from Eq. [6.1] is added into the Lagrangian for our Gibbs function. If ions are present, the charge constraint [6.2] is also put in. Therefore, our Lagrangian is:

$$\mathcal{L} = \sum_{j=1}^{\text{NS}} \mu_j \mathcal{N}_j + \sum_{i=1}^{\text{NE}} \lambda_i \left[\sum_{j=1}^{\text{NS}} a_{ij}\mathcal{N}_j - b_i^\circ\right] + \lambda_q \left[\sum_{j=0}^{\text{NS}} q_j \mathcal{N}_j\right] \tag{6.7}$$

The contraints available for the Gibbs energy are:

- Hold temperature (T) and pressure (P) constant.
- Hold enthalpy (H) and pressure (P) constant.
- Hold entropy (S) and pressure (P) constant.

For all Gibbs minimizations, we need the molar constraint:

$$\sum_{j=1}^{\text{NS}} \mathcal{N}_j - \mathcal{N} = 0 \tag{6.8}$$

For (TP) minimization, only the elemental and charge (if present) constraints are necessary. For (HP) minimization, the constraint is defined by:

$$h' - h'_0 = 0 \tag{6.9}$$

Where h'_0 is the specified enthalpy in J/kg, and h' is solved for during the minimization process as:

$$h' = \sum_{j=1}^{\text{NS}} \mathcal{N}_j H_j^\circ \tag{6.10}$$

For (SP) minimization, the constraint is defined by:

$$s' - s'_0 = 0 \tag{6.11}$$

Where s'_0 is the specified entropy and s' is solved for in the minimization process as:

$$s' = \sum_{j=1}^{\text{NS}} \mathcal{N}_j S_j \tag{6.12}$$

where:

$$S_j = S_j^\circ - \hat{R} \ln \frac{\mathcal{N}_j}{\mathcal{N}} - \hat{R} \ln p \tag{6.13}$$

6.1.6 Derivatives

We now take the derivatives of $\mathcal{L}$ wrt $\mathcal{N}_j$, and each λ, denoting them as f:

$$f_1 = \frac{\partial \mathcal{L}}{\partial \mathcal{N}_j} = \mu_j + \sum_{i=1}^{\text{NE}} \lambda_i a_{ij} + \lambda_q q_j = 0 \tag{6.14}$$

$$f_2 = \frac{\partial \mathcal{L}}{\partial \lambda_i} = \sum_{i=1}^{\text{NS}} a_{ij} \mathcal{N}_j - b_i = 0 \tag{6.15}$$

$$f_3 = \frac{\partial \mathcal{L}}{\partial \lambda_q} = \sum_{j=0}^{\text{NS}} q_j \mathcal{N}_j = 0 \tag{6.16}$$

These three equations are our minimization functions. Through use of the aforementioned Newton-Raphson method, we can solve these equations. Additional residual equations for total moles of the system, enthalpy, and entropy are:

$$f_4 = \sum_{j=1}^{\text{NS}} \mathcal{N}_j - \mathcal{N} = 0 \tag{6.17}$$

$$f_5 = \frac{h' - h'_0}{\hat{R}T} = \frac{\sum_{j=1}^{\text{NS}} \mathcal{N}_j H_j^\circ - h'_0}{\hat{R}T} \tag{6.18}$$

$$f_6 = \frac{s' - s'_0}{\hat{R}} = \frac{\sum_{j=1}^{\text{NS}} \mathcal{N}_j S_j - s'_0}{\hat{R}} \tag{6.19}$$

Special care needs to be taken in the enthalpy constraint to include the enalpies of formation and/or reference enthalpies, if applicable.

6.1.7 Derivatives of f_1

First, we must do some algebraic manipulation. We start by expanding Eq. [6.14] with Eq. [6.6]:

$$\frac{\partial \mathcal{L}}{\partial \mathcal{N}_j} = \mu_j^\circ + \hat{R}T\left[\ln\left(\frac{p}{p_{\text{ref}}}\right) + \ln\left(\frac{\mathcal{N}_j}{\mathcal{N}}\right)\right] + \sum_{i=1}^{\text{NE}} \lambda_i a_{ij} + \lambda_q q_j = 0 \tag{6.20}$$

Dividing through by $\hat{R}T$ and setting $\pi = -\lambda/\hat{R}T$ gives:

$$f_1 = \frac{\mu_j^\circ}{\hat{R}T} + \ln\frac{p}{p_{\text{ref}}} + ln\mathcal{N}_j - \ln\mathcal{N} - \sum_{i=1}^{\text{NE}} \pi_i a_{i,j} - \pi_q q_j = 0 \tag{6.21}$$

We now need to take the partial derivatives of our functions wrt our non-linear variables as well as each π, and then multiple it by the correction variables. You can easily take derviatives wrt $\ln x$ by converting from x to $\exp(\ln x)$.

$$\frac{\partial f_1}{\partial[\ln\mathcal{N}_j]}\Delta\ln\mathcal{N}_j = \Delta\ln\mathcal{N}_j \tag{6.22}$$

$$\frac{\partial f_1}{\partial[\ln\mathcal{N}]}\Delta\ln\mathcal{N} = -\Delta\ln\mathcal{N} \tag{6.23}$$

$$\frac{\partial f_1}{\partial[\ln T]}\Delta\ln T = -\frac{H_j^\circ}{\hat{R}T}\Delta\ln T \tag{6.24}$$

$$\frac{\partial f_1}{\partial[\pi_i]}\pi_i = -a_{ij}\pi_i \tag{6.25}$$

$$\frac{\partial f_1}{\partial[\pi_q]}\pi_q = -q_j\pi_q \tag{6.26}$$

Combining these into the form of Eq. [6.4] gives

$$\Delta\ln\mathcal{N}_j - \Delta\ln\mathcal{N} - \sum_{i=1}^{\text{NE}} a_{ij}\pi_i - q_j\pi_q - \frac{H_j^\circ}{\hat{R}T}\Delta\ln T = \mu_j + \sum_{i=1}^{\text{NE}} \pi_i a_{ij} + \pi_q q_j \tag{6.27}$$

From before, the argument is made that $\pi_i = 0$ at the start of every iteration, so this equation becomes:

$$\boxed{\Delta\ln\mathcal{N}_j - \Delta\ln\mathcal{N} - \sum_{i=1}^{\text{NE}} a_{ij}\pi_i - q_j\pi_q - \frac{H_j^\circ}{\hat{R}T}\Delta\ln T = -\frac{\mu_j}{\hat{R}T}}$$

There will be one of these equation for each species.

6.1.8 Derivatives of f_2

$$\frac{\partial f_2}{\partial[\ln\mathcal{N}_j]}\Delta\ln\mathcal{N}_j = a_{ij}\mathcal{N}_j\Delta\ln\mathcal{N}_j \tag{6.28}$$

$$\frac{\partial f_2}{\partial[\ln\mathcal{N}]}\Delta\ln\mathcal{N} = \frac{\partial f_2}{\partial[\pi_i]}\pi_i = \frac{\partial f_2}{\partial[\ln T]} = \frac{\partial f_2}{\partial[\pi_q]}\pi_q = 0 \tag{6.29}$$

Which gives:

$$\boxed{\sum_{j=1}^{\text{NS}} a_{ij}\mathcal{N}_j \Delta \ln \mathcal{N}_j = b_i^\circ - \sum_{j=1}^{\text{NS}} a_{ij}\mathcal{N}_j}$$

6.1.9 Derivatives of f_3

$$\frac{\partial f_3}{\partial[\ln \mathcal{N}_j]} \Delta \ln \mathcal{N}_j = q_j \mathcal{N}_j \Delta \ln \mathcal{N}_j \tag{6.30}$$

$$\frac{\partial f_3}{\partial[\ln \mathcal{N}]} \Delta \ln \mathcal{N} = \frac{\partial f_3}{\partial[\ln T]} = \frac{\partial f_3}{\partial[\pi_i]}\pi_i = \frac{\partial f_3}{\partial[\pi_q]}\pi_q = 0 \tag{6.31}$$

Giving:

$$\boxed{\sum_{j=1}^{\text{NS}} q_j \mathcal{N}_j \Delta \ln \mathcal{N}_j = -\sum_{j=1}^{\text{NS}} q_i \mathcal{N}_j}$$

6.1.10 Derivatives of f_4

$$\frac{\partial f_4}{\partial[\ln \mathcal{N}_j]} \Delta \ln \mathcal{N}_j = \mathcal{N}_j \Delta \ln \mathcal{N}_j \tag{6.32}$$

$$\frac{\partial f_4}{\partial[\ln \mathcal{N}]} \Delta \ln \mathcal{N} = -\mathcal{N} \Delta \ln \mathcal{N} \tag{6.33}$$

$$\frac{\partial f_4}{\partial[\ln T]} = \frac{\partial f_4}{\partial[\pi_i]}\pi_i = \frac{\partial f_4}{\partial[\pi_q]}\pi_q = 0 \tag{6.34}$$

$$\boxed{\sum_{j=1}^{\text{NS}} \mathcal{N}_j \Delta \ln \mathcal{N}_j - \mathcal{N} \Delta \ln \mathcal{N} = \mathcal{N} - \sum_{j=1}^{\text{N}} \mathcal{N}_j} \tag{6.35}$$

6.1.11 Derivatives of f_5

$$\frac{\partial f_5}{\partial[\ln \mathcal{N}_j]} \Delta \ln \mathcal{N}_j = \mathcal{N}_j \frac{H_j^\circ}{\hat{R}T} \Delta \ln \mathcal{N}_j \tag{6.36}$$

$$\frac{\partial f_5}{\partial[\ln T]} = \sum_{j=1}^{\text{NS}} \mathcal{N}_j \frac{c_{p,j}^\circ}{\hat{R}} \Delta \ln T \tag{6.37}$$

$$\frac{\partial f_5}{\partial[\ln \mathcal{N}]} \Delta \ln \mathcal{N} = \frac{\partial f_5}{\partial[\pi_i]}\pi_i = \frac{\partial f_5}{\partial[\pi_q]}\pi_q = 0 \tag{6.38}$$

$$\boxed{\sum_{j=1}^{\text{NS}} \mathcal{N}_j \frac{H_j^\circ}{\hat{R}T} \Delta \ln \mathcal{N}_j + \sum_{j=1}^{\text{NS}} \mathcal{N}_j \frac{c_{p,j}^\circ}{\hat{R}} \Delta \ln T = \frac{h' - h_0'}{\hat{R}T}}$$

6.1.12 Derivatives of f_6

$$\frac{\partial f_6}{\partial[\ln \mathcal{N}_j]}\Delta \ln \mathcal{N}_j = \mathcal{N}_j \frac{S_j}{\hat{R}}\Delta \ln \mathcal{N}_j \tag{6.39}$$

$$\frac{\partial f_6}{\partial[\ln T]} = \sum_{j=1}^{\text{NS}} \mathcal{N}_j \frac{c^\circ_{p,j}}{\hat{R}}\Delta \ln T \tag{6.40}$$

$$\frac{\partial f_6}{\partial[\ln \mathcal{N}]}\Delta \ln \mathcal{N} = \frac{\partial f_6}{\partial[\pi_i]}\pi_i = \frac{\partial f_6}{\partial[\pi_q]}\pi_q = 0 \tag{6.41}$$

$$\boxed{\sum_{j=1}^{\text{NS}} \mathcal{N}_j \frac{S_j}{\hat{R}}\Delta \ln \mathcal{N}_j + \sum_{j=1}^{\text{NS}} \mathcal{N}_j \frac{c^\circ_{p,j}}{\hat{R}}\Delta \ln T = \frac{s'_0 - s'}{\hat{R}} + \mathcal{N} - \sum_{j=1}^{\text{NS}} \mathcal{N}_j}$$

All six of the equations are now listed for convenience:

$$\Delta \ln \mathcal{N}_j - \Delta \ln \mathcal{N} - \sum_{i=1}^{\text{NE}} a_{ij}\pi_i - q_j \pi_q - \frac{H^\circ_j}{\hat{R}T}\Delta \ln T = -\frac{\mu_j}{\hat{R}T} \tag{6.42}$$

$$\sum_{j=1}^{\text{NS}} \mathcal{N}_j \Delta \ln \mathcal{N}_j - \mathcal{N}\Delta \ln \mathcal{N} = \mathcal{N} - \sum_{j=1}^{\text{NS}} \mathcal{N}_j \tag{6.43}$$

$$\sum_{j=1}^{\text{NS}} a_{ij}\mathcal{N}_j \Delta \ln \mathcal{N}_j = b^\circ_i - \sum_{j=1}^{\text{NS}} a_{ij}\mathcal{N}_j \tag{6.44}$$

$$\sum_{j=1}^{\text{NS}} q_j \mathcal{N}_j \Delta \ln \mathcal{N}_j = -\sum_{j=1}^{\text{NS}} q_i \mathcal{N}_j \tag{6.45}$$

$$\sum_{j=1}^{\text{NS}} \mathcal{N}_j \frac{H^\circ_j}{\hat{R}T}\Delta \ln \mathcal{N}_j + \sum_{j=1}^{\text{NS}} \mathcal{N}_j \frac{c^\circ_{p,j}}{\hat{R}}\Delta \ln T = \frac{h'_0 - h'}{\hat{R}T} \tag{6.46}$$

$$\sum_{j=1}^{\text{NS}} \mathcal{N}_j \frac{S_j}{\hat{R}}\Delta \ln \mathcal{N}_j + \sum_{j=1}^{\text{NS}} \mathcal{N}_j \frac{c^\circ_{p,j}}{\hat{R}}\Delta \ln T = \frac{s'_0 - s'}{\hat{R}} + \mathcal{N} - \sum_{j=1}^{\text{NS}} \mathcal{N}_j \tag{6.47}$$

6.1.13 Reduced Gibbs Equations

A shortcut can be made to make this system of equations smaller. This is achieved by solving Eq. [6.42] for $\Delta \ln \mathcal{N}_j$ and pluggin it into Eqs. [6.43] - [6.47]

$$\Delta \ln \mathcal{N}_j = \Delta \ln \mathcal{N} + \sum_{i=1}^{\text{NE}} a_{ij}\pi_i - \frac{\mu_j}{\hat{R}T} + q_j \pi_q + \frac{H^\circ_j}{\hat{R}T}\Delta \ln T \tag{6.48}$$

Substitution into Eqs. [6.44] - [6.47] yields:

$$\sum_{i=1}^{\text{NE}} \left[\sum_{j=1}^{\text{NS}} a_{kj}a_{ij}\mathcal{N}_j\right]\pi_i + \left[\sum_{j=1}^{\text{NS}} a_{kj}\mathcal{N}_j\right]\Delta \ln \mathcal{N} + \left[\sum_{j=1}^{\text{NS}} a_{kj}q_j\mathcal{N}_j\right]\pi_q + \left[\sum_{j=1}^{\text{NS}} a_{kj}\mathcal{N}_j \frac{H^\circ_j}{\hat{R}T}\right]\Delta \ln T$$
$$= b^\circ_k - \sum_{j=1}^{\text{NS}} a_{kj}\mathcal{N}_j + \sum_{j=1}^{\text{NS}} a_{kj}\mathcal{N}_j \frac{\mu_j}{\hat{R}T} \tag{6.49}$$

$$\sum_{i=1}^{\mathrm{NE}}\left[\sum_{j=1}^{\mathrm{NS}} a_{ij}\mathcal{N}_j\right]\pi_i + \left[\sum_{j=1}^{\mathrm{NS}}\mathcal{N}_j - \mathcal{N}\right]\Delta\ln\mathcal{N} + \left[\sum_{j=1}^{\mathrm{NS}} q_j\mathcal{N}_j\right]\pi_q + \left[\sum_{j=1}^{\mathrm{NS}}\mathcal{N}_j\frac{H_j^\circ}{\hat{R}T}\right]\Delta\ln T$$
$$= \mathcal{N} - \sum_{j=1}^{\mathrm{NS}}\mathcal{N}_j + \sum_{j=1}^{\mathrm{NS}}\mathcal{N}_j\frac{\mu_j}{\hat{R}T} \tag{6.50}$$

$$\sum_{i=1}^{\mathrm{NE}}\left[\sum_{j=1}^{\mathrm{NS}} a_{ij}q_j\mathcal{N}_j\right]\pi_i + \left[\sum_{j=1}^{\mathrm{NS}} q_j\mathcal{N}_j\right]\Delta\ln\mathcal{N} + \left[\sum_{j=1}^{\mathrm{NS}} q_j^2\mathcal{N}_j\right]\pi_q + \left[\sum_{j=1}^{\mathrm{NS}} q_j\mathcal{N}_j\frac{H_j^\circ}{\hat{R}T}\right]\Delta\ln T$$
$$= \sum_{j=1}^{\mathrm{NS}} q_j\mathcal{N}_j\frac{\mu_j}{\hat{R}T} - \sum_{j=1}^{\mathrm{NS}} q_j\mathcal{N}_j \tag{6.51}$$

$$\sum_{i=1}^{\mathrm{NE}}\left[\sum_{j=1}^{\mathrm{NS}} a_{ij}\mathcal{N}_j\frac{H_j^\circ}{\hat{R}T}\right]\pi_i + \left[\sum_{j=1}^{\mathrm{NS}}\mathcal{N}_j\frac{H_j^\circ}{\hat{R}T}\right]\Delta\ln\mathcal{N} + \left[\sum_{j=1}^{\mathrm{NS}} q_j\mathcal{N}_j\frac{H_j^\circ}{\hat{R}T}\right]\pi_q$$
$$+ \left[\sum_{j=1}^{\mathrm{NS}}\mathcal{N}_j\left(\frac{H_j^\circ}{\hat{R}T}\right)^2 + \sum_{j=1}^{\mathrm{NS}}\mathcal{N}_j\frac{c_{p,j}^\circ}{\hat{R}}\right]\Delta\ln T = \frac{h_0' - h'}{\hat{R}T} + \sum_{j=1}^{\mathrm{NS}}\mathcal{N}_j\frac{H_j^\circ}{\hat{R}T}\frac{\mu_j}{\hat{R}T} \tag{6.52}$$

$$\sum_{i=1}^{\mathrm{NE}}\left[\sum_{j=1}^{\mathrm{NS}} a_{ij}\mathcal{N}_j\frac{S_j}{\hat{R}}\right]\pi_i + \left[\sum_{j=1}^{\mathrm{NS}}\mathcal{N}_j\frac{S_j}{\hat{R}}\right]\Delta\ln\mathcal{N} + \left[\sum_{j=1}^{\mathrm{NS}} q_j\mathcal{N}_j\frac{S_j}{\hat{R}}\right]\pi_q$$
$$+ \left[\sum_{j=1}^{\mathrm{NS}}\mathcal{N}_j\frac{S_j}{\hat{R}}\frac{H_j^\circ}{\hat{R}T} + \sum_{j=1}^{\mathrm{NS}}\mathcal{N}_j\frac{c_{p,j}^\circ}{\hat{R}}\right]\Delta\ln T = \frac{s_0' - s'}{\hat{R}} + \mathcal{N} - \sum_{j=1}^{\mathrm{NS}}\mathcal{N}_j + \sum_{j=1}^{\mathrm{NS}}\mathcal{N}_j\frac{S_j}{\hat{R}}\frac{\mu_j}{\hat{R}T} \tag{6.53}$$

There are NE number of Eq. [6.50], and only one of all others. Once this system has been solved, you recover $\mathcal{N}_j$ with Eq. [6.48].

The equations are colored according to when they are needed. Red coloring denotes terms that arise from the elemental constraint condition. These are included no matter what. Blue denotes terms that are added when you include the charge constraint. Violet terms are added for both HP and SP constraints. orange and teal are the rows added for enthalpy or entropy constraints, respectively.

6.1.14 Helmholtz Minimization

The Helmholtz energy is defined as:

$$F = G - pV \tag{6.54}$$

Where G is the Gibbs free energy, p is the pressure, and V is the volume. Substitution of G yields:

$$F = \sum_{j=1}^{\text{NS}} \mu_j \mathcal{N}_j - pV \tag{6.55}$$

Where the chemical potential is redefined as:

$$\mu_j = \mu_j^\circ + \hat{R}T \ln\left(\frac{\mathcal{N}_j R' T}{V}\right) \tag{6.56}$$

and $R' = \hat{R} \cdot 10^{-5}$ is the same as dividing $\hat{R}$ by the reference pressure of 1 bar.

6.1.15 Constraints

The contraints available for the Helmholtz energy are:

- Hold temperature (T) and volume (V) constant.
- Hold internal energy (U) and volume (V) constant.
- Hold entropy (S) and volume (V) constant.

For (TV) minimization, only the elemental and charge (if present) constraints are necessary. For (UV) minimization, the constraint is defined by:

$$u' - u'_0 = 0 \tag{6.57}$$

Where u'_0 is the specified internal energy, and u' is solved for during the minimization process as:

$$u' = \sum_{j=1}^{\text{NS}} \mathcal{N}_j U_j^\circ \tag{6.58}$$

As stated before, special care needs to be taken to include the enthalpies of formation and reference enthalpies if applicable. For (SV) minimization, the constraint is defined by:

$$s' - s'_0 = 0 \tag{6.59}$$

Where s'_0 is the specified entropy and s' is solved for during the minimization process as:

$$s' = \sum_{j=1}^{\text{NS}} \mathcal{N}_j S_j \tag{6.60}$$

where:

$$S_j = S_j^\circ - \hat{R} \ln \mathcal{N}_j - \hat{R} \ln\left(\frac{R'T}{V}\right) \tag{6.61}$$

The Lagrangian for the Helmholtz energy minimization process is then defined as:

$$\mathcal{L} = \sum_{j=1}^{\text{NS}} \mu_j \mathcal{N}_j - pV + \sum_{i=1}^{\text{NE}} \lambda_i \left[\sum_{j=1}^{\text{NS}} a_{ij} \mathcal{N}_j - b_i^\circ\right] + \lambda_q \left[\sum_{j=1}^{\text{NS}} q_j \mathcal{N}_j\right] \tag{6.62}$$

6.1.16 Derivatives

We first find our functions f that are equal to 0 by taking the derivatives wrt $\mathcal{N}_j$, λ_i, and λ_q:

$$f_1 = \frac{\partial \mathcal{L}}{\partial \mathcal{N}_j} = \mu_j + \sum_{i=1}^{\mathrm{NE}} \lambda_i a_{ij} + \lambda_q q_j = 0 \tag{6.63}$$

$$f_2 = \frac{\partial \mathcal{L}}{\partial \lambda_i} = \sum_{j=1}^{\mathrm{NS}} a_{ij}\mathcal{N}_j - b_i^\circ = 0 \tag{6.64}$$

$$f_3 = \frac{\partial \mathcal{L}}{\partial \lambda_q} = \sum_{j=1}^{\mathrm{NS}} q_j\mathcal{N}_j - 0 = 0 \tag{6.65}$$

We now form the second derivatives that define our Jacobian in the Newton step. We expand the chemical potential term using Eq. [6.56] and non-dimensionalize the equation by $\hat{R}T$. We also set $\pi = -\lambda/\hat{R}T$.

6.1.17 Derivatives for f_1

The derivatives for f_1 are then:

$$\frac{\partial f_1}{\partial[\ln \mathcal{N}_j]}\Delta \ln \mathcal{N}_j = \Delta \ln \mathcal{N}_j \tag{6.66}$$

$$\frac{\partial f_1}{\partial[\ln T]}\Delta \ln T = -\frac{U_j^\circ}{\hat{R}T}\ln T \tag{6.67}$$

$$\frac{\partial f_1}{\partial \pi_i}\pi_i = -a_{ij}\pi_i \tag{6.68}$$

$$\frac{\partial f_1}{\partial \pi_q}\pi_q = -q_j\pi_q \tag{6.69}$$

The derivation for Eq. [6.67] is carried out in more depth in section [6.4]. Therefore, our final result is:

$$\boxed{\Delta \ln \mathcal{N}_j - \sum_{i=1}^{\mathrm{NE}} a_{ij}\pi_i - q_j\pi_q - \frac{U_j^\circ}{\hat{R}T}\Delta \ln T = -\frac{\mu_j}{\hat{R}T}}$$

There will be NS number of these equations in our system.

6.1.18 Derivatives for f_2

Next, we take the derivatives of f_2:

$$\frac{\partial f_2}{\partial[\ln \mathcal{N}_j]}\Delta \ln \mathcal{N}_j = \mathcal{N}_j a_{i,j}\Delta \ln \mathcal{N}_j \tag{6.70}$$

$$\frac{\partial f_2}{\partial[\ln T]}\Delta \ln T = \frac{\partial f_2}{\partial \pi_i}\pi_i = \frac{\partial f_2}{\partial \pi_q}\pi_q = 0 \tag{6.71}$$

$$\boxed{\sum_{j=1}^{\text{NS}} \mathcal{N}_j a_{ij} \Delta \ln \mathcal{N}_j = b_i^\circ - \sum_{j=1}^{\text{NS}} a_{ij} \mathcal{N}_j}$$

There will be NE of these equations.

6.1.19 Derivatives for f_3

For f_3 we get:

$$\frac{\partial f_3}{\partial [\ln \mathcal{N}_j]} \Delta \ln \mathcal{N}_j = \mathcal{N}_j q_j \Delta \ln \mathcal{N}_j \tag{6.72}$$

$$\frac{\partial f_3}{\partial [\ln T]} \Delta \ln T = \frac{\partial f_3}{\partial \pi_i} \pi_i = \frac{\partial f_3}{\partial \pi_q} \pi_q = 0 \tag{6.73}$$

$$\boxed{\sum_{j=1}^{\text{NS}} \mathcal{N}_j q_j \Delta \ln \mathcal{N}_j = -\sum_{j=1}^{\text{NS}} q_j \mathcal{N}_j}$$

6.1.20 Derivatives for f_4

For internal energy, we have:

$$f_4(\ln \mathcal{N}_j, \ln T) = \sum_{j=1}^{\text{NS}} \mathcal{N}_j U_j^\circ - u_0' = u' - u_0' \tag{6.74}$$

Using identity [6.134] helps with the temperature derivative. The derivatives are then:

$$\frac{\partial f_4}{\partial [\ln \mathcal{N}_j]} \Delta \ln \mathcal{N}_j = \frac{\mathcal{N}_j U_j^\circ}{\hat{R} T} \Delta \ln \mathcal{N}_j \tag{6.75}$$

$$\frac{\partial f_4}{\partial [\ln T]} \Delta \ln T = \sum_{j=1}^{\text{NS}} [T \mathcal{N}_j c_{v,j}^\circ] \Delta \ln T \tag{6.76}$$

$$\frac{\partial f_4}{\partial \pi_i} \pi_i = \frac{\partial f_4}{\partial \pi_q} \pi_q = 0 \tag{6.77}$$

After non-dimensionalizing, we are left with:

$$\boxed{\sum_{j=1}^{\text{NS}} \frac{\mathcal{N}_j U_j^\circ}{\hat{R} T} \Delta \ln \mathcal{N}_j + \sum_{j=1}^{\text{NS}} \frac{\mathcal{N}_j c_{v,j}^\circ}{R} \Delta \ln T = \frac{u_0' - u'}{\hat{R} T}}$$

6.1.21 Derivatives for f_5

For the entropy constraint we have:

$$f_5(\ln \mathcal{N}_j, \ln T) = \sum_{j=1}^{\text{NS}} \mathcal{N}_j S_j - s_0' = s' - s_0' \tag{6.78}$$

Again, identity [6.134] assists us. The derivates are:

$$\frac{\partial f_5}{\partial[\ln \mathcal{N}_j]}\Delta \ln \mathcal{N}_j = \mathcal{N}_j[S_j - \hat{R}]\Delta \ln \mathcal{N}_j \tag{6.79}$$

$$\frac{\partial f_5}{\partial[\ln T]}\Delta \ln T = \mathcal{N}_j c^\circ_{v,j}\Delta \ln T \tag{6.80}$$

$$\frac{\partial f_5}{\partial \pi_i}\pi_i = \frac{\partial f_5}{\partial \pi_q}\pi_q = 0 \tag{6.81}$$

Which, after non-dimensionalizing gives the singular equation:

$$\boxed{\sum_{j=1}^{\text{NS}} \mathcal{N}_j \left[\frac{S_j}{\hat{R}} - 1\right]\Delta \ln \mathcal{N}_j + \mathcal{N}_j \frac{c^\circ_{v,j}}{\hat{R}}\Delta \ln T = \frac{s'_0 - s'}{\hat{R}}} \tag{6.82}$$

For convenience, they are all listed here together:

$$\Delta \ln \mathcal{N}_j - \sum_{i=1}^{\text{NE}} a_{ij}\pi_i - q_j\pi_q - \frac{U^\circ_j}{\hat{R}T}\Delta \ln T = -\frac{\mu_j}{\hat{R}T} \tag{6.83}$$

$$\sum_{j=1}^{\text{NS}} \mathcal{N}_j a_{ij}\Delta \ln \mathcal{N}_j = b^\circ_i - \sum_{j=1}^{\text{NS}} a_{ij}\mathcal{N}_j \tag{6.84}$$

$$\sum_{j=1}^{\text{NS}} \mathcal{N}_j q_j \Delta \ln \mathcal{N}_j = -\sum_{j=1}^{\text{NS}} q_j \mathcal{N}_j \tag{6.85}$$

$$\sum_{j=1}^{\text{NS}} \frac{\mathcal{N}_j U^\circ_j}{\hat{R}T}\Delta \ln \mathcal{N}_j + \sum_{j=1}^{\text{NS}} \frac{\mathcal{N}_j c^\circ_{v,j}}{R}\Delta \ln T = \frac{u'_0 - u'}{\hat{R}T} \tag{6.86}$$

$$\sum_{j=1}^{\text{NS}} \mathcal{N}_j \left[\frac{S_j}{\hat{R}} - 1\right]\Delta \ln \mathcal{N}_j + \mathcal{N}_j \frac{c^\circ_{v,j}}{\hat{R}}\Delta \ln T = \frac{s'_0 - s'}{\hat{R}} \tag{6.87}$$

6.1.22 Reduced Helmholtz Equations

A "simplification" can be made to the above system of equations by solving Eq. [6.83] for $\Delta \ln \mathcal{N}_j$:

$$\Delta \ln \mathcal{N}_j = \sum_{i=1}^{\text{NE}} a_{i,j}\pi_i - \frac{\mu_j}{\hat{R}T} + q_j\pi_q + \frac{U^\circ_j}{\hat{R}T}\Delta \ln T \tag{6.88}$$

By substituting this into Eqs. [6.84] - [6.87], we can shrink the system of equations dramatically. The resulting set of equations is:

$$\sum_{i=1}^{\text{NE}} \left[\sum_{j=1}^{\text{NS}} a_{kj}a_{ij}\mathcal{N}_j\right]\pi_i + \left[\sum_{j=1}^{\text{NS}} a_{kj}q_j\mathcal{N}_j\right]\pi_q + \left[\sum_{j=1}^{\text{NS}} a_{kj}\mathcal{N}_j \frac{U^\circ_j}{\hat{R}T}\right]\Delta \ln T$$
$$= b^\circ_k - \sum_{j=1}^{\text{NS}} a_{kj}\mathcal{N}_j + \sum_{j=1}^{\text{NS}} a_{kj}\mathcal{N}_j \frac{\mu_j}{\hat{R}T} \tag{6.89}$$

$$\sum_{i=1}^{NE}\left[\sum_{j=1}^{NS} a_{ij} q_j \mathcal{N}_j\right]\pi_i + \left[\sum_{j=1}^{NS} q_j^2 \mathcal{N}_j\right]\pi_q + \left[\sum_{j=1}^{NS} q_j \mathcal{N}_j \frac{U_j^\circ}{\hat{R}T}\right]\Delta \ln T = \sum_{j=1}^{NS} q_j \mathcal{N}_j \frac{\mu_j}{\hat{R}T} - \sum_{j=1}^{NS} q_j \mathcal{N}_j \tag{6.90}$$

$$\sum_{i=1}^{NE}\left[\sum_{j=1}^{NS} a_{ij} \mathcal{N}_j \frac{U_j^\circ}{\hat{R}T}\right]\pi_i + \left[\sum_{j=1}^{NS} q_j \mathcal{N}_j \frac{U_j^\circ}{\hat{R}T}\right]\pi_q + \left[\sum_{j=1}^{NS} \mathcal{N}_j \left(\frac{U_j^\circ}{\hat{R}T}\right)^2 + \sum_{j=1}^{NS} \mathcal{N}_j \frac{c_{v,j}^\circ}{\hat{R}}\right]\Delta \ln T = \frac{u_0' - u'}{\hat{R}T} + \sum_{j=1}^{NS} \mathcal{N}_j \frac{U_j^\circ}{\hat{R}T}\frac{\mu_j}{\hat{R}T} \tag{6.91}$$

$$\sum_{i=1}^{NE}\left[\sum_{j=1}^{NS} a_{ij} \mathcal{N}_j \left(\frac{S_j}{\hat{R}} - 1\right)\right]\pi_i + \left[\sum_{j=1}^{NS} q_j \mathcal{N}_j \left(\frac{S_j}{\hat{R}} - 1\right)\right]\pi_q + \left[\sum_{j=1}^{NS} \mathcal{N}_j \left(\frac{S_j}{\hat{R}} - 1\right)\frac{U_j^\circ}{\hat{R}T} + \sum_{j=1}^{NS} \mathcal{N}_j \frac{c_{v,j}^\circ}{\hat{R}}\right]\Delta \ln T = \frac{s_0' - s'}{\hat{R}} + \sum_{j=1}^{NS} \mathcal{N}_j \frac{\mu_j}{\hat{R}T}\left(\frac{S_j}{\hat{R}} - 1\right) \tag{6.92}$$

There will be NE number of Eqs. [6.89], and one of [6.90] if charge is added, and one of [6.91]/[6.92] depending on which one you made need.

The equations are colored according to when they are needed. Red coloring denotes terms that arise from the elemental constraint condition. These are included no matter what. Blue denotes terms that are added when you include the charge constraint. Violet terms are added for both UV and SV constraints. orange and teal are the rows added for either internal energy or entropy constraints, respectively.

6.2 Thermodynamic Derivatives

This section goes over computation of thermodynamic derivatives for finding things like c_p, c_v, a, etc.

6.2.1 Base equations and derivatives:

We start with some important equations that are necessary for future differentiation. The equations shown below are (in order of appearance) the equation of state, the Lagrangian derivatives previously denoted as f_1, f_2, and f_3, and the total mole constraint:

$$pV = \mathcal{N}RT \tag{6.93}$$

$$f_1 = \frac{\mu_j^\circ}{\hat{R}T} + \ln\frac{p}{p_{\text{ref}}} + ln\mathcal{N}_j - \ln\mathcal{N} - \sum_{i=1}^{NE}\pi_i a_{i,j} - \pi_q q_j = 0 \tag{6.94}$$

$$f_2 = \sum_{j=1}^{\text{NS}} a_{ij}\mathcal{N}_j - b_i = 0 \tag{6.95}$$

$$f_3 = \sum_{j=0}^{\text{NS}} q_j\mathcal{N}_j = 0 \tag{6.96}$$

$$f_4 = \sum_{j}^{\text{NS}} \mathcal{N}_j - \mathcal{N} = 0 \tag{6.97}$$

Taking the natural log of each side of Eq. [6.93] gives:

$$\ln V = \ln \mathcal{N} + \ln R + \ln T - \ln p \tag{6.98}$$

Certain important derivatives can be found by differentiating this equation wrt [$\ln p$] holding T constant, or [$\ln T$] holding p constant:

$$\left.\frac{\partial \ln V}{\partial \ln T}\right|_p = 1 + \left.\frac{\partial \ln \mathcal{N}}{\partial \ln T}\right|_p \tag{6.99a}$$
$$\left.\frac{\partial \ln V}{\partial \ln p}\right|_T = \left.\frac{\partial \ln \mathcal{N}}{\partial \ln p}\right|_T - 1 \tag{6.99b}$$

A third fundemental equation is that of the specific heat c_p. Specific terms will be derived later, but it has the form:

$$\frac{c_{p,e}}{\hat{R}} = \sum_{j=1}^{\text{NS}} \mathcal{N}_j \frac{H_j^\circ}{\hat{R}T}\left(\frac{\partial \ln \mathcal{N}_j}{\partial \ln T}\right)_p + \sum_{j=1}^{\text{NS}} \mathcal{N}_j \frac{c_{p,j}^\circ}{\hat{R}} \tag{6.100}$$

Expanding for the $\ln \mathcal{N}_j$ derivative (seen below) in yields:

$$\begin{aligned}\frac{c_{p,e}}{\hat{R}} = \sum_{i=1}^{\text{NE}} \left[\sum_{j=1}^{\text{NS}} a_{ij}\mathcal{N}_j \frac{H_j^\circ}{\hat{R}T}\right]\left(\frac{\partial \pi_i}{\partial \ln T}\right)_p + \left[\sum_{j=1}^{\text{NS}} \mathcal{N}_j \frac{H_j^\circ}{\hat{R}T}\right]\left(\frac{\partial \ln \mathcal{N}}{\partial \ln T}\right)_p \\ + \left[\sum_{j=1}^{\text{NS}} q_j\mathcal{N}_j \frac{H_j^\circ}{\hat{R}T}\right]\left(\frac{\partial \pi_q}{\partial \ln T}\right)_p + \sum_{j=1}^{\text{NS}} \mathcal{N}_j \left(\frac{H_j^\circ}{\hat{R}T}\right)^2 + \sum_{j=1}^{\text{NS}} \mathcal{N}_j \frac{c_{p,j}^\circ}{\hat{R}}\end{aligned} \tag{6.101}$$

Where the blue term is only used if ionization is present. These three derivatives are the core derivatives for the use of the *Bridgman tables*. All thermodynamic derivatives can be formed with the two logarithmic derivatives and c_p.

6.2.2 Notes on derivatives and Bridgmann tables

An extremely important relationship is found between logarithmic derivatives and linear derivatives. We first start with the chain rule of differentiation:

$$\frac{\partial \ln x}{\partial \ln y} = \frac{\partial(\ln x)}{\partial x}\frac{\partial x}{\partial y}\frac{\partial y}{\partial(\ln y)} \tag{6.102}$$

Since the first dervative on the RHS comes out to $1/x$, and the third dervative is just y, this gives:

$$\frac{\partial \ln x}{\partial \ln y} = \frac{y}{x}\frac{\partial x}{\partial y} \tag{6.103}$$

This identity can be used alongside the Bridggman tables, which will be used excessively in the next few sections. A copy of the Bridgman tables is contained in the `documents` directory of this repository. We can start with an example, and then future derivatives will follow the same procedure, but will not be detailed.

The speed of sound can be expressed as:

$$a^2 = \left(\frac{\partial p}{\partial \rho}\right)_S = \frac{p}{\rho}\left(\frac{\partial \ln p}{\partial \ln \rho}\right)_s = -\frac{p}{\rho}\left(\frac{\partial \ln p}{\partial \ln V}\right)_S \tag{6.104}$$

We ignore the fraction coefficient and just put the derivative into Bridgman form as:

$$\left(\frac{\partial \ln p}{\partial \ln V}\right)_S = \frac{V}{p}\frac{(\partial p)_S}{(\partial V)_S} \tag{6.105}$$

The Bridgmann tables give:

$$(\partial p)_S = -(\partial S)_p = -\frac{c_p}{T} \tag{6.106}$$

$$(\partial V)_S = -(\partial S)_V = \frac{1}{T}\left[c_p \left.\frac{\partial V}{\partial p}\right|_T + T\left(\frac{\partial V}{\partial T}\right)_p^2\right] \tag{6.107}$$

Putting the derivatives inside the second equation into logarithmic form gives:

$$(\partial V)_S = -\frac{1}{T}\left[\frac{c_p V}{P}\left(\frac{\partial \ln V}{\partial \ln p}\right)_T + \frac{V^2}{T}\left(\frac{\partial \ln V}{\partial \ln T}\right)_p^2\right] \tag{6.108}$$

Therefore, the final result for the derivative is:

$$\left(\frac{\partial \ln p}{\partial \ln V}\right)_S = \frac{c_p}{c_p\left(\frac{\partial \ln V}{\partial \ln p}\right)_T + \frac{PV}{T}\left(\frac{\partial \ln V}{\partial \ln T}\right)_p^2} \tag{6.109}$$

Which is exactly the same result given in Gordon and Mcbride. As you can see, this derivative uses the three fundemental derivatives derived in the previous section. Therefore, we need to solve for these three derivatives first in order to find speed of sound or other derivatives needed for CFD.

6.2.3 Temperature derivatives and c_p

We can now go through equations $f_1 - f_4$ and take derivatives wrt $[\ln T]$ holding p constant:

$$\left[\frac{\partial f_1}{\partial \ln T}\right]_p = \left[\frac{\partial \ln \mathcal{N}_j}{\partial \ln T}\right]_p - \left[\frac{\partial \ln \mathcal{N}}{\partial \ln T}\right]_p - \sum_{i=1}^{\mathrm{NE}} a_{ij}\left[\frac{\partial \pi_i}{\partial \ln T}\right]_p - q_j\left[\frac{\partial \pi_q}{\partial \ln T}\right]_p = \frac{H_j^\circ}{\hat{R}T} \tag{6.110}$$

$$\left[\frac{\partial f_2}{\partial \ln T}\right]_p = \sum_{j=1}^{\text{NS}} a_{ij}\mathcal{N}_j \left[\frac{\partial \ln \mathcal{N}_j}{\partial \ln T}\right]_p = 0 \tag{6.111}$$

$$\left[\frac{\partial f_3}{\partial \ln T}\right]_p = \sum_{j}^{\text{NS}} q_j\mathcal{N}_j \left[\frac{\partial \ln \mathcal{N}_j}{\partial \ln T}\right]_p = 0 \tag{6.112}$$

$$\sum_{j}^{\text{NS}} \mathcal{N}_j \left[\frac{\partial \ln \mathcal{N}_j}{\partial \ln T}\right]_p - \mathcal{N}\left[\frac{\partial \ln \mathcal{N}}{\partial \ln T}\right]_p = 0 \tag{6.113}$$

This system of equations allows us to solve for the terms in brackets which are needed to recover our three fundemental derivatives. However, as before, we look to create a reduced set of equations. We do so by solving Eq. [6.110] for $\partial \ln \mathcal{N}_j / \partial \ln T|_p$ and substituting this result into the rest of the equations. Since the majority of the terms are always used, only the ionization term is added if necessary, so it is colored blue to show this.

$$\left[\frac{\partial \ln \mathcal{N}_j}{\partial \ln T}\right]_p = \left[\frac{\partial \ln \mathcal{N}}{\partial \ln T}\right]_p + \sum_{i=1}^{\text{NE}} a_{ij}\left[\frac{\partial \pi_i}{\partial \ln T}\right]_p + q_j\left[\frac{\partial \pi_q}{\partial \ln T}\right]_p + \frac{H_j^\circ}{\hat{R}T} \tag{6.114}$$

The resulting reduced equations are then:

$$\sum_{i=1}^{\text{NE}}\left[\sum_{j=1}^{\text{NS}} a_{kj}a_{ij}\mathcal{N}_j\right]\left(\frac{\partial \pi_i}{\partial \ln T}\right)_p + \left[\sum_{j=1}^{\text{NS}} a_{kj}\mathcal{N}_j\right]\left(\frac{\partial \ln \mathcal{N}}{\partial \ln T}\right)_p + \left[\sum_{j=1} a_{kj}q_j\mathcal{N}_j\right]\left(\frac{\partial \pi_q}{\partial \ln T}\right)_p = -\sum_{j=1}^{\text{NS}} a_{kj}\mathcal{N}_j\frac{H_j^\circ}{\hat{R}T} \tag{6.115}$$

$$\sum_{i=1}^{\text{NE}}\left[\sum_{j=1}^{\text{NS}} a_{ij}\mathcal{N}_j\right]\left(\frac{\partial \pi_i}{\partial \ln T}\right)_p + \left[\sum_{j=1}^{\text{NS}} q_j\mathcal{N}_j\right]\left(\frac{\partial \pi_q}{\partial \ln T}\right)_p = -\sum_{j=1}^{\text{NS}} \mathcal{N}_j\frac{H_j^\circ}{\hat{R}T} \tag{6.116}$$

$$\sum_{i=1}^{\text{NE}}\left[\sum_{j=1}^{\text{NS}} a_{ij}q_j\mathcal{N}_j\right]\left(\frac{\partial \pi_i}{\partial \ln T}\right)_p + \left[\sum_{j=1}^{\text{NS}} q_j\mathcal{N}_j\right]\left(\frac{\partial \ln \mathcal{N}}{\partial \ln T}\right)_p + \left[\sum_{j=1}^{\text{NS}} q_j^2\mathcal{N}_j\right]\left(\frac{\partial \pi_q}{\partial \ln T}\right)_p = -\sum_{j=1}^{\text{NS}} q_j\mathcal{N}_j\frac{H_j^\circ}{\hat{R}T} \tag{6.117}$$

Since this is done after the Newton solve, an iterative procedure is not used. This also gets rid of a term inside the final equation, since $\sum_{j=1}^{\text{NS}} \mathcal{N}_j - \mathcal{N}$ is 0. There will be NE number of the first equation, and 1 of the other two equations. From our Newton iteration to solve for $\mathcal{N}_j$, we can assume that the difference between the penultimate and final iteration is negligible, so we can actually use a number of terms from the original Jacobian that has already been set up. We also dont even need to recover the $\mathcal{N}_j$ derivative wrt $\ln T$, we can just plug those results directly into the above equation for the specific heat,

c_p.

Solving this system of equations gives us the ability to solve for Eq. [6.99a] and c_p. We now just need to solve for the final fundemental derivative.

6.2.4 Pressure derivatives

We now look at the derivatives of the functions f_i wrt $[\ln p]$ holding T constant. This gives the following unreduced system:

$$\left[\frac{\partial \ln \mathcal{N}_j}{\partial \ln p}\right]_T - \left[\frac{\partial \ln \mathcal{N}}{\partial \ln p}\right]_T - \sum_{i=1}^{\text{NE}} a_{ij}\left[\frac{\partial \pi_i}{\partial \ln p}\right]_T - q_j\left[\frac{\partial \pi_q}{\partial \ln p}\right]_T = -1 \tag{6.118}$$

$$\sum_{j=1}^{\text{NS}} a_{ij}\mathcal{N}_j\left[\frac{\partial \ln \mathcal{N}_j}{\partial \ln p}\right]_T = 0 \tag{6.119}$$

$$\sum_{j=1}^{\text{NS}} q_j\mathcal{N}_j\left[\frac{\partial \ln \mathcal{N}_j}{\partial \ln p}\right]_T = 0 \tag{6.120}$$

$$\sum_{j}^{\text{NS}} \mathcal{N}_j\left[\frac{\partial \ln \mathcal{N}_j}{\partial \ln p}\right]_T - \mathcal{N}\left[\frac{\partial \ln \mathcal{N}}{\partial \ln p}\right]_T = 0 \tag{6.121}$$

We then solve Eq. [6.118] for the $\mathcal{N}_j$ derivative:

$$\left[\frac{\partial \ln \mathcal{N}_j}{\partial \ln p}\right]_T = \left[\frac{\partial \ln \mathcal{N}}{\partial \ln p}\right]_T + \sum_{i=1}^{\text{NE}} a_{ij}\left[\frac{\partial \pi_i}{\partial \ln p}\right]_T + q_j\left[\frac{\partial \pi_q}{\partial \ln p}\right]_T - 1 \tag{6.122}$$

Substituting the resultant RHS into the rest of the equations to get the reduced equations:

$$\sum_{i=1}^{\text{NE}}\left[\sum_{j=1}^{\text{NS}} a_{kj}a_{ij}\mathcal{N}_j\right]\left(\frac{\partial \pi_i}{\partial \ln p}\right)_T + \left[\sum_{j=1}^{\text{NS}} a_{kj}\mathcal{N}_j\right]\left(\frac{\partial \ln \mathcal{N}}{\partial \ln p}\right)_T + \left[\sum_{j=1}^{\text{NS}} a_{kj}q_j\mathcal{N}_j\right]\left(\frac{\partial \pi_q}{\partial \ln p}\right)_T = \sum_{j=1}^{\text{NS}} a_{kj}\mathcal{N}_j \tag{6.123}$$

$$\sum_{i=1}^{\text{NE}}\left[\sum_{j=1}^{\text{NS}} a_{ij}\mathcal{N}_j\right]\left(\frac{\partial \pi_i}{\partial \ln p}\right)_T + \left[\sum_{j=1}^{\text{NS}} q_j\mathcal{N}_j\right]\left(\frac{\partial \pi_q}{\partial \ln p}\right)_T = \sum_{j=1}^{\text{NS}} \mathcal{N}_j \tag{6.124}$$

$$\sum_{i=1}^{\text{NE}}\left[\sum_{j=1}^{\text{NS}} a_{ij}q_j\mathcal{N}_j\right]\left(\frac{\partial \pi_i}{\partial \ln p}\right)_T + \left[\sum_{j=1}^{\text{NS}} q_j\mathcal{N}_j\right]\left(\frac{\partial \ln N}{\partial \ln p}\right)_T + \left[\sum_{j=1}^{\text{NS}} q_j^2 N_j\right]\left(\frac{\partial \pi_q}{\partial \ln p}\right)_T = \sum_{j=1}^{\text{NS}} q_j\mathcal{N}_j \tag{6.125}$$

Solving this system of equations is again, a non-iterative procedure, and we can use the previous

Jacobian since the results are the same. Only the RHS changes. This makes the most expensive part just the LU solve. We can than find c_v as:

$$\frac{c_v}{\hat{R}} = \frac{c_p}{\hat{R}} + \frac{p}{\rho \hat{R} T} \left[\frac{\left(\frac{\partial \ln V}{\partial \ln T}\Big|_p \right)^2}{\frac{\partial \ln V}{\partial \ln p}\Big|_T} \right] \tag{6.126}$$

And then the final fundemental derivative is also found from Eq. [6.99b].

6.2.5 CFD derivatives

The two necessary derivatives for CFD are:

$$\frac{\partial p}{\partial \rho}\Big|_e \qquad \frac{\partial p}{\partial e}\Big|_\rho \tag{6.127}$$

This can be found with the Bridggman tables. First, we focus on the derivative wrt density. Let us express it with specific volume so that we can use Bridggman tables:

$$\frac{\partial p}{\partial \rho}\Big|_e = \frac{\partial p}{\partial V}\Big|_e \frac{\partial V}{\partial \rho}\Big|_e = -\frac{1}{\rho^2} \frac{\partial p}{\partial V}\Big|_e = -V^2 \frac{\partial p}{\partial V}\Big|_e \tag{6.128}$$

Tables, with logarithmic derivatives yield:

$$\frac{\partial p}{\partial V}\Big|_e = \frac{-c_p + \frac{pV}{T}\left[\frac{\partial \ln V}{\partial \ln T}\right]_p}{-\frac{c_p V}{p}\left[\frac{\partial \ln V}{\partial \ln p}\right]_T - \frac{V^2}{T}\left[\frac{\partial \ln V}{\partial \ln T}\right]_p^2} \tag{6.129}$$

$$\frac{\partial p}{\partial \rho}\Big|_e = -V^2 \frac{\partial p}{\partial V}\Big|_e = \frac{-c_p + \frac{pV}{T}\left[\frac{\partial \ln V}{\partial \ln T}\right]_p}{\frac{c_p}{pV}\left[\frac{\partial \ln V}{\partial \ln p}\right]_T + \frac{1}{T}\left[\frac{\partial \ln V}{\partial \ln T}\right]_p^2} \tag{6.130}$$

$$\boxed{\frac{\partial p}{\partial \rho}\Big|_e = \frac{p}{\rho} \frac{\left[\frac{\partial \ln V}{\partial \ln T}\right]_p - \frac{c_p}{\hat{R}}}{\left[\frac{\partial \ln V}{\partial \ln T}\right]_p^2 + \frac{c_p}{\hat{R}}\left[\frac{\partial \ln V}{\partial \ln p}\right]_T}} \tag{6.131}$$

The derivative with respect to internal energy is:

$$\frac{\partial p}{\partial e}\Big|_\rho = \frac{\partial p}{\partial e}\Big|_V \tag{6.132}$$

Tables yield:

$$\boxed{\frac{\partial p}{\partial e}\Big|_\rho = \frac{-\rho\left[\frac{\partial \ln V}{\partial \ln T}\right]_p}{\left[\frac{\partial \ln V}{\partial \ln T}\right]_p^2 + \frac{c_p}{R}\left[\frac{\partial \ln V}{\partial \ln p}\right]_T}} \tag{6.133}$$

Note the common denominators in each equation.

6.3 Standard State Chemical Potential Derivatives

Some important derivatives are derived here tha are used in the matrix equations for Gibbs and Helmholtz energy minimization. First, we state a chain rule identity that is very useful:

$$\frac{\partial}{\partial T} = \frac{\partial[\ln(T)]}{\partial T}\frac{\partial}{\partial[\ln(T)]} \rightarrow \frac{\partial}{\partial[\ln(T)]} = T\frac{\partial}{\partial T} \tag{6.134}$$

Then, taking the derivative of μ_j^0 gives:

$$\frac{\partial}{\partial[\ln(T)]}\left(\frac{\mu_j^0}{RT}\right) = T\frac{\partial}{\partial T}\left(\frac{\mu_j^0}{RT}\right) \tag{6.135}$$

We expand μ_j^0/RT as one of the following two equation depending on if we are using Gibbs or Helmholtz minimization:

$$\frac{\mu_j^0}{RT} = \frac{H_j^0}{RT} - \frac{S_j^0}{R} \tag{6.136}$$

$$\frac{\mu_j^0}{RT} = \frac{U_j^0}{RT} - \frac{S_j^0}{R} + 1 \tag{6.137}$$

6.3.1 Useful Relationships

$$\frac{c_v^\circ}{\hat{R}} = \frac{c_{p,j}^\circ}{\hat{R}} - 1 \tag{6.138}$$

$$\frac{\mu_j^\circ}{\hat{R}T} = \frac{H_j^\circ}{\hat{R}T} - \frac{S_j^\circ}{\hat{R}} \tag{6.139}$$

$$\frac{U_j^\circ}{\hat{R}T} = \frac{H_j^\circ}{\hat{R}T} - 1 \tag{6.140}$$

6.3.2 Gibbs Derivative

First, we take the derivative of [6.136] for Gibbs minimization.

$$T\frac{\partial}{\partial T}\left(\frac{H_j^0}{RT} - \frac{S_j^0}{R}\right) = T\left(\frac{\partial_T(H_j^0)\cdot RT - H_j^0\cdot R}{(RT)^2} - \frac{\partial_T(S_j^0)}{R}\right) \tag{6.141}$$

Here, ∂_T denotes $(\partial/\partial T)$. If:

$$c_{p,j}^0 = \frac{\partial H_j^0}{\partial T}, \text{ and } \frac{c_{p,j}^0}{T} = \frac{\partial S_j^0}{\partial T} \tag{6.142}$$

This reduces to:

$$T\left[\frac{c_{p,j}^0}{RT} - \frac{H_j^0}{RT^2} - \frac{c_{p,j}^0}{RT}\right] = -\frac{H_j^0}{RT} \tag{6.143}$$

Therefore:

$$\boxed{\frac{\partial}{\partial[\ln(T)]}\left(\frac{\mu_j^0}{RT}\right) = -\frac{H_j^0}{RT}} \tag{6.144}$$

6.4 Helmholtz Derivative

Using equation [6.137] gives us:

$$T\frac{\partial}{\partial T}\left(\frac{U_j^0}{RT}+1-\frac{S_j^0}{R}\right)=T\left(\frac{\partial_T(U_j^0)\cdot RT-U_j^0\cdot R}{(RT)^2}-\frac{\partial_T(S_j^0)}{R}\right) \tag{6.145}$$

Recalling $\partial_T(S_j^0)$ from equation [6.142] and using:

$$c_{v,j}^0=\frac{\partial U_j^0}{\partial T} \tag{6.146}$$

Equation [6.145] reduced to:

$$T\left[\frac{c_{v,j}^0}{RT}-\frac{U_j^0}{RT^2}-\frac{c_{p,j}^0}{RT}\right] \tag{6.147}$$

Since:

$$\frac{c_{v,j}^0}{R}-\frac{c_{p.j}^0}{R}=-1 \tag{6.148}$$

The final derivative is:

$$\boxed{\frac{\partial}{\partial[\ln(T)]}\left(\frac{\mu_j^0}{RT}\right)=-\frac{U_j^0}{RT}-1} \tag{6.149}$$

7 Thermochemical Nonequilibrium CFD

7.1 Species conservation equation

$$\underbrace{\frac{\partial}{\partial t}\rho_s}_{1} + \underbrace{\frac{\partial}{\partial x_j}\rho_s u_j}_{2} = \underbrace{\frac{\partial}{\partial x_j}(\rho D_s \frac{\partial}{\partial x_j} y_s)}_{3} + \underbrace{\dot{\omega}_s}_{4} \tag{7.1}$$

The terms are:

1. Rate of change of mass of species s.
2. Flux of mass due to convection.
3. Molecular diffusion flux of mass.
4. Chemical production rate.

The variables are:

- ρ_s is the density of species s
- u_j is the mass-averaged velocity in the j-direction
- D_s is the effective diffusion coefficient of species s
- y_s is the molar fraction of species s
- $\dot{\omega}_s$ is the chemical production rate of species s

7.2 Total continuity equation

For chemically reacting flows, the total continuity equation is unaffected. It is:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \tag{7.2}$$

Since no mass can be created or destroyed, the condition of atomic conservation is applied, which yields:

$$\sum_s \dot{w}_s = 0 \tag{7.3}$$

7.3 Mixture momentum conservation

$$\underbrace{\frac{\partial}{\partial t}\rho u_i}_{1} + \underbrace{\frac{\partial}{\partial x_j}\rho u_i u_j}_{2} = -\underbrace{\frac{\partial p}{\partial x_i}}_{3} + \underbrace{\frac{\partial}{\partial x_j}\left[\mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) - \frac{2}{3}\mu\frac{\partial u_k}{\partial x_k}\delta_{ij}\right]}_{4} \tag{7.4}$$

The terms are:

1. Rate of change of ith component of momentum.
2. Flux of ith component of momentum due to convection.

3. Pressure forces in ith direction.

4. Viscous terms acting in ith direction.

Here, the mixture pressure is defined with Dalton's law of partial pressures, and the pressure of an individual heavy species is:

$$p_s = \frac{\rho_s \hat{R} T}{\mathcal{M}_s} \tag{7.5}$$

For electron pressure, we get:

$$p_e = \frac{\rho_e \hat{R} T_e}{\mathcal{M}_e} \tag{7.6}$$

Where the e subscript denotes an electron property.

7.4 Vibrational energy conservation

$$\begin{aligned}\underbrace{\frac{\partial}{\partial t}\rho e_v}_{1} + \underbrace{\frac{\partial}{\partial x_j}\rho e_v u_j}_{2} = \underbrace{\frac{\partial}{\partial x_j}\left(\eta_v \frac{\partial T_v}{\partial x_j}\right)}_{3} + \underbrace{\frac{\partial}{\partial x_j}\left(\rho \sum_{s=1}^{\text{NS}} h_{v,s} D_s \frac{\partial y_s}{\partial x_j}\right)}_{4} \\ + \underbrace{\sum_{\text{s = mol.}} \rho_s \frac{(e^*_{v,s} - e_{v,s})}{< \tau_s >}}_{5} + \underbrace{\sum_{\text{s=mol.}} \rho_s \frac{(e^{**}_{v,s} - e_{v,s})}{< \tau_{es} >}}_{6} + \underbrace{\sum_{\text{s=mol.}} \dot{w}_s \hat{D}_s}_{7}\end{aligned} \tag{7.7}$$

The terms are:

1. Rate of change of vibrational energy per unit volume.

2. The flux of vibrational energy due to convection.

3. The conduction of vibrational energy due to vibrational temperature gradients.

4. The diffusion of vibrational energy due to molecular concentration gradients.

5. The energy exchange (relaxation) between vibrational and translational modes due to collisions within the cell.

6. The energy exchange between vibrational and electronic modes.

7. The vibrational lost/gained due to molecular depletion (dissociation) or production (recombination).

The variables are:

- e_v is the vibrational energy per unit mass, defined as $e_v = \sum_{s=1}^{\text{NS}} \frac{\rho_s e_{v,s}}{\rho}$. e_v is defined as a function of T_v, the vibrational temperature.

- η_v is the vibrational thermal conductivity.

- $h_{v,s}$ is the vibrational enthalpy. It is identical to $e_{v,s}$.
- $e^{*}_{v,s}$ is the vibrational energy of species s at the translational-rotational temperature.
- $e^{**}_{v,s}$ is the vibrational energy at the electron temperature.
- $< \tau_s >$ is the characteristic relaxation times for translational-vibrational (T-V) energy exchange.
- $< \tau_{es} >$ is the characteristic relation times for electron-vibrational (e-V) energy exchange.
- $\hat{D}_s$ is the vibration energy level representative of those molecules of species s which are created or destroyed (recombined or dissociated) because of their high vibrational quantum numbers.

7.5 Electron and electronic excitation energy conservation

$$\underbrace{\frac{\partial}{\partial t}\rho e_e}_{1} + \underbrace{\frac{\partial}{\partial x_j}\left[u_j(\rho e_e + p_e)\right]}_{2} = \underbrace{u_j\frac{\partial p_e}{\partial x_j}}_{3} + \underbrace{\frac{\partial}{\partial x_j}\left(\eta_e\frac{\partial T_e}{\partial x_j}\right)}_{4} + \underbrace{\frac{\partial}{\partial x_j}\left(\rho\sum_{s=1}^{\text{NS}} h_{e,s}D_s\frac{\partial y_s}{\partial x_j}\right)}_{5}$$

$$+ \underbrace{2\rho_e\frac{3}{2}\hat{R}(T-T_e)\sum_{s=1}^{\text{NS - 1}}\frac{\nu_{es}}{\mathcal{M}_s}}_{6} - \underbrace{\sum_{\text{s=ion}}\dot{n}_{e,s}\hat{I}_s}_{7} - \underbrace{\sum_{\text{s=mol.}}\rho_s\frac{e^{**}_{v,s}-e_{v,s}}{<\tau_{es}>}}_{8} - \underbrace{Q_{\text{rad}}}_{9} \quad (7.8)$$

Terms are:

1. Rate of change of electronic energy per unit volume.
2. The flux of electronic enthalpy due to convection.
3. The word done on electrons by an electron field by an electron pressure gradient.
4. The conduction of electronic energy due to electron temperature gradient.
5. The diffuction of electronic energy due to concentration gradients.
6. The energy exchange due to elastic collisions between electrons and heavy particles.
7. Energy loss due to electron impact ionization (note the sum indices sum over ionized species only).
8. The relaxation due to inelastic collisions between electrons and molecules.
9. Rate of energy loss due to radiation caused by electronic transitions.

The variables are:

- e_e is the electronic energy per unit mass and contains contributions from electronic energy levels. It is defined as $e_e = \sum_{s=1}^{\text{NS}}\frac{\rho_s e_{e,s}}{\rho}$
- $e_{e,s}$ is the electronic energy per unit mass of species s and is defined as a function of T_e.
- T_e is the electron temperature.

- η_e is the is the electronic thermal conductivity.
- $h_{e,s}$ is the electronic enthalpy per unit mass and is idential to $e_{e,s}$ for all species except free electrons. For free electrons it is defined as: $h_{e,e} = e_{e,e} + \dfrac{\hat{R}T_e}{\mathcal{M}_e}$
- $\nu_{e,s}$ is the effective collision frequency of electrons with heavy particles.
- $\dot{n}_{e,s}$ is the molar rate of production of species s by electron impact ionizations.
- $\hat{I}_s$ is the energy per unit mole lost by a free electron in producing species s through electron impact ionization.
- Q_{rad} is the radiant energy transfer rate due to electron transitions.

7.6 Total energy conservation

$$\underbrace{\frac{\partial}{\partial t}\rho E}_{1} + \underbrace{\frac{\partial}{\partial x_j}\rho H u_j}_{2} = \underbrace{\frac{\partial}{\partial x_j}\left(\eta\frac{\partial T}{\partial x_j} + \eta_v\frac{\partial T_v}{\partial x_j} + \eta_e\frac{\partial T_e}{\partial x_j}\right)}_{3} + \underbrace{\frac{\partial}{\partial x_j}\left(\rho\sum_{s=1}^{\text{NS}} h_s D_s\frac{\partial y_s}{\partial x_j}\right)}_{4} + \underbrace{\frac{\partial}{\partial x_j}\left[u_i\mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)\right]}_{5} - \underbrace{Q_{\text{rad}}}_{6} \tag{7.9}$$

The terms are:

1. Rate of change of total energy per unit volume.
2. The flux of total enthalpy due to convection.
3. The conduction of energy due to temperature, vibrational temperature, and electron temperature gradients.
4. The diffusion of energy due to concentration gradients.
5. The word done by shear forces.
6. The rate of energy loss due to radiation caused by electronic transitions.

The variables are:

- E is the total energy per unit volume, defined as $E = \dfrac{u_i u_i}{2} + \displaystyle\sum_{s=1}^{\text{NS}} \frac{\rho_s e_s}{\rho}$
- H is the total enthalpy per unit mass, defined as $H = E + \dfrac{p}{\rho}$
- η is the frozen thermal conductivity of heavy particles. The part in which exchanges of translational and rotational energy occur.

7.7 Two temperature model

Everything before this assumes a three-temperature model, where there is a single temperature dedicated to translational-rotational energy, vibrational energy, and electronic-electron energies. The two temperature model assumes that one temperature can be used for describing the energy of translational-rotational energies, and then the rest (vib-electronic-electron) are desibed by the second temperature, T_V. Therefore:

$$T_v = T_e = T_V \tag{7.10}$$

By combining the vibrational and electronic energy equations, we get the vibrational-electronic energy conservation equation:

$$\begin{aligned}
\underbrace{\frac{\partial}{\partial t}\rho e_V}_{1} + \underbrace{\frac{\partial}{\partial x_j}\rho e_V u_j}_{2} &= \underbrace{-p_e\frac{\partial u_j}{\partial x_j}}_{3} + \underbrace{\frac{\partial}{\partial x_j}\left[(\eta_v + \eta_e)\frac{\partial T_V}{\partial x_j}\right]}_{4} \\
&+ \underbrace{\frac{\partial}{\partial x_j}\left(\rho\sum_{s=1}^{\text{NS}} h_{V,s} D_s \frac{\partial y_s}{\partial x_j}\right)}_{5} + \underbrace{\sum_{s=mol.}\rho_s\frac{(e^*_{v,s} - e_{v,s})}{<\tau_s>}}_{6} \\
&+ \underbrace{2\rho_e\frac{3}{2}\hat{R}(T - T_V)\sum_{s=1}^{\text{NS - 1}}\frac{\nu_{es}}{\mathcal{M}_s}}_{7} - \underbrace{\sum_{\text{s=ion}}\dot{n}_{e,s}\hat{I}_s}_{8} + \underbrace{\sum_{s=mol.}\dot{w}_s\hat{D}_s}_{9} - \underbrace{Q_{\text{rad}}}_{10}
\end{aligned} \tag{7.11}$$

Where:

$$e_V = e_v + e_e \tag{7.12}$$

$$h_{V,s} = h_{v,s}h_{e,s} \tag{7.13}$$

7.8 Derivatives

7.8.1 Partial derivatives of pressure (P)

$$\frac{\partial p}{\partial \rho_s} = \left(\frac{\hat{R}}{\mathcal{M}_s} - c_{vs}\frac{R}{c_v}\right)T + \frac{R}{c_v}\left[\frac{1}{2}(u^2 + v^2 + w^2) - h_s^\circ\right] \tag{7.14a}$$

$$\frac{\partial p}{\partial(\rho u)} = -u\frac{R}{c_v} \tag{7.14b}$$

$$\frac{\partial p}{\partial(\rho v)} = -v\frac{R}{c_v} \tag{7.14c}$$

$$\frac{\partial p}{\partial(\rho w)} = -w\frac{R}{c_v} \tag{7.14d}$$

$$\frac{\partial p}{\partial E_v} = -\frac{R}{c_v} \tag{7.14e}$$

$$\frac{\partial p}{\partial E} = \frac{R}{c_v} \tag{7.14f}$$

7.8.2 Partial derivatives of total energy (E)

$$\frac{\partial E}{\partial \rho_s} = \left(c_{vs} - \frac{c_v}{R} \frac{\hat{R}}{\mathcal{M}_s} \right) T + \frac{1}{2}(u^2 + v^2 + w^2) + e_v + h_s^\circ \tag{7.15a}$$

$$\frac{\partial E}{\partial u} = \rho u \tag{7.15b}$$

$$\frac{\partial E}{\partial v} = \rho v \tag{7.15c}$$

$$\frac{\partial E}{\partial w} = \rho w \tag{7.15d}$$

$$\frac{\partial E}{\partial e_v} = \sum_{s=1}^{\mathrm{NS}} \rho_s \tag{7.15e}$$

$$\frac{\partial E}{\partial p} = \frac{c_v}{R} \tag{7.15f}$$

7.8.3 Partial derivatives of T

$$\frac{\partial T}{\partial \rho_s} = \frac{1}{\rho c_v} \left(-c_{vs} T + \frac{1}{2}(u^2 + v^2 + w^2) - h_s^\circ \right) \tag{7.16a}$$

$$\frac{\partial T}{\partial (\rho u)} = -\frac{u}{\rho c_v} \tag{7.16b}$$

$$\frac{\partial T}{\partial (\rho v)} = -\frac{v}{\rho c_v} \tag{7.16c}$$

$$\frac{\partial T}{\partial (\rho w)} = -\frac{w}{\rho c_v} \tag{7.16d}$$

$$\frac{\partial T}{\partial E_v} = -\frac{1}{\rho c_v} \tag{7.16e}$$

$$\frac{\partial T}{\partial E} = \frac{1}{\rho c_v} \tag{7.16f}$$

$$\frac{\partial c_r}{\partial \rho_s} = (\delta_{rs} - c_r)\frac{1}{\rho} \tag{7.16g}$$

7.8.4 Partial derivatives of vibrational temperature (T_v)

$$\frac{\partial T_v}{\partial \rho_i} = -e_{vi} \Big/ \sum_{s=1}^{\mathrm{NS}} \left[\rho_s \left(\frac{\partial e_{vs}}{\partial T_v} \right) \right] \tag{7.17a}$$

$$\frac{\partial T_v}{\partial (\rho u)} = 0 \tag{7.17b}$$

$$\frac{\partial T_v}{\partial (\rho v)} = 0 \tag{7.17c}$$

$$\frac{\partial T_v}{\partial (\rho w)} = 0 \tag{7.17d}$$

$$\frac{\partial T_v}{\partial E_v} = 1 \Big/ \sum_{s=1}^{\mathrm{NS}} \left[\rho_s \left(\frac{\partial e_{vs}}{\partial T_v} \right) \right] \tag{7.17e}$$

$$\frac{\partial T_v}{\partial E} = 0 \tag{7.17f}$$

7.8.5 Partial derivatives of vibrational energy ($e_v s$)

$$\frac{\partial e_{vs}}{\partial \rho_i} = \left(\frac{\partial e_{vs}}{\partial T_v}\right)_{T_v} \frac{\partial T_v}{\partial \rho_i}, \qquad \frac{\partial e^*_{vs}}{\partial \rho_i} = \left(\frac{\partial e_{vs}}{\partial T_v}\right)_{T_v=T} \frac{\partial T_v}{\partial \rho_i} \tag{7.18a}$$

$$\frac{\partial e_{vs}}{\partial (\rho u)} = \left(\frac{\partial e_{vs}}{\partial T_v}\right)_{T_v} \frac{\partial T_v}{\partial (\rho u)}, \qquad \frac{\partial e^*_{vs}}{\partial (\rho u)} = \left(\frac{\partial e_{vs}}{\partial T_v}\right)_{T_v=T} \frac{\partial T_v}{\partial (\rho u)} \tag{7.18b}$$

$$\frac{\partial e_{vs}}{\partial (\rho v)} = \left(\frac{\partial e_{vs}}{\partial T_v}\right)_{T_v} \frac{\partial T_v}{\partial (\rho v)}, \qquad \frac{\partial e^*_{vs}}{\partial (\rho v)} = \left(\frac{\partial e_{vs}}{\partial T_v}\right)_{T_v=T} \frac{\partial T_v}{\partial (\rho v)} \tag{7.18c}$$

$$\frac{\partial e_{vs}}{\partial (\rho w)} = \left(\frac{\partial e_{vs}}{\partial T_v}\right)_{T_v} \frac{\partial T_v}{\partial (\rho w)}, \qquad \frac{\partial e^*_{vs}}{\partial (\rho w)} = \left(\frac{\partial e_{vs}}{\partial T_v}\right)_{T_v=T} \frac{\partial T_v}{\partial (\rho w)} \tag{7.18d}$$

$$\frac{\partial e_{vs}}{\partial E_v} = \left(\frac{\partial e_{vs}}{\partial T_v}\right)_{T_v} \frac{\partial T_v}{\partial E_v}, \qquad \frac{\partial e^*_{vs}}{\partial E_v} = \left(\frac{\partial e_{vs}}{\partial T_v}\right)_{T_v=T} \frac{\partial T_v}{\partial E_v} \tag{7.18e}$$

$$\frac{\partial e_{vs}}{\partial E} = \left(\frac{\partial e_{vs}}{\partial T_v}\right)_{T_v} \frac{\partial T_v}{\partial E}, \qquad \frac{\partial e^*_{vs}}{\partial E} = \left(\frac{\partial e_{vs}}{\partial T_v}\right)_{T_v=T} \frac{\partial T_v}{\partial E} \tag{7.18f}$$

7.9 Jacobians

7.9.1 Flux Jacobians

7.9.2 Source term Jacobians - Third body reactions

As stated, for third body reactions, the Arrhenius rate coefficient k_f can be expressed as an efficiency times a base rate coefficient that is independant of the third body:

$$k_f(T) = \epsilon_i \gamma(T) \tag{7.19}$$

where

$$\gamma(T) = C_c T^\eta e^{-E_a/T} \tag{7.20}$$

For a third-body reaction, the rate of production of species s can be expressed as:

$$\frac{d[X_i]}{dt} = (\nu''_i - \nu'_i) \underbrace{\sum_{i=1}^{\text{NS}} \epsilon_i [M_i]}_{E_{\text{eff}}} \overbrace{\left\{ \underbrace{\gamma(T_f)[X_{s,f}]^{\nu'_s}}_{F} - \underbrace{\frac{\gamma(T_b)}{K_c(T_b)}[X_{s,b}]^{\nu''_s}}_{B} \right\}}^{L_r} \tag{7.21}$$

Another useful equation for the derivations in the future is:

$$R_r = \underbrace{\sum_{i=1}^{\text{NS}} \epsilon_i [M_i]}_{E_{\text{eff}}} \overbrace{\left\{ \underbrace{\gamma(T_f)[X_{s,f}]^{\nu'_s}}_{F} - \underbrace{\frac{\gamma(T_b)}{K_c(T_b)}[X_{s,b}]^{\nu''_s}}_{B} \right\}}^{L_r} \tag{7.22}$$

Here, T_f is the temperature used to compute the forward rate coefficient, and T_b is the temperature used to compute the backward rate coefficient. For now, we are only interest in the two underbracketed sections, E_{eff} and L_r. The reaction rate (without the stoichiometric coefficients in the front) can then be described by:

$$\mathcal{R}_r = E_{\text{eff}} L_r \tag{7.23}$$

The partial derivative of this reaction rate with respect to some conserved variable U_j is then:

$$\frac{\partial \mathcal{R}_r}{\partial U_j} = \frac{\partial E_{\text{eff}}}{\partial U_j} L_r + E_{\text{eff}} \frac{\partial L_r}{\partial U_j} \tag{7.24}$$

The two derivatives on the right hand side are then:

$$\frac{\partial E_{\text{eff}}}{\partial U_j} = \begin{cases} \frac{\epsilon_j}{\mathcal{M}_j}, & \text{if } U_j = \rho_j \\ 0, & \text{if } U_j = \rho u, \rho v, \text{etc.} \end{cases} \tag{7.25}$$

$$\begin{aligned} \frac{\partial L_r}{\partial U_j} = & \overbrace{\left\{ \frac{\partial \gamma(T_f)}{\partial U_j} \right\} \left[\frac{\rho_{s,f}}{\mathcal{M}_{s,f}} \right]^{\nu_s'} + \gamma(T_f) \left\{ \frac{\partial}{\partial U_j} \left[\frac{\rho_{s,b}}{\mathcal{M}_{s,b}} \right]^{\nu_s'} \right\}}^{\partial \mathrm{F}} \\ & \underbrace{- \left\{ \frac{\partial}{\partial U_j} \left[\frac{\gamma(T_b)}{K_c(T_b)} \right] \right\} \left[\frac{\rho_{s,b}}{\mathcal{M}_{s,b}} \right]^{\nu_s''} + \frac{\gamma(T_b)}{K_c(T_b)} \left\{ \frac{\partial}{\partial U_j} \left[\frac{\rho_{s,b}}{\mathcal{M}_{s,b}} \right]^{\nu_s''} \right\}}_{\partial \mathrm{B}} \end{aligned} \tag{7.26}$$

Note that the product across concentrations has been removed in the L_r derivative. This is because of the form of third-body reactions and how it has gotten moved into the E_{eff} term.

Taking it step by step, we first look at three important derivatives. Since $\gamma(T_f)$ is a function of the forward rate-controlling temperature, which can have a form of $T^q T_v^{1-q}$, while $\gamma(T_b)$ only depends on T or T_e. We will use the chain rule to compute the derivative of $\gamma(T_c)$, where T_c is the rate controlling temperature for forwards or backwards rates:

$$T = T_{tr}^q T_v^{1-q} T_e^p \tag{7.27}$$

$$\frac{\partial \gamma(T_c)}{\partial U_j} = \frac{\partial \gamma(T_f)}{\partial T} \frac{\partial T}{\partial U_j} = \frac{\partial \gamma(T_f)}{\partial T_c} \frac{\partial T_c}{\partial T} \frac{\partial T}{\partial U_j} \tag{7.28}$$

$$= \left[\frac{\partial}{\partial T_c} \left(C_c T_f^\eta e^{-E_a/T_c} \right) \right] \frac{\partial T_c}{\partial T} \frac{\partial T}{\partial U_j} \tag{7.29}$$

$$= \left[\eta C_c T_c^{\eta-1} e^{-E_a/T_c} + C_c T_c^\eta e^{-E_a/T_c} E_a T_c^{-2} \right] \frac{\partial T_c}{\partial T} \frac{\partial T}{\partial U_j} \tag{7.30}$$

$$= \gamma(T_c) \left[\frac{\eta}{T_c} + \frac{E_a}{T_c^2} \right] \left[\frac{\partial T_c}{\partial T} \frac{\partial T}{\partial U_j} + \frac{\partial T_c}{\partial T_v} \frac{\partial T_v}{\partial U_j} \right] \tag{7.31}$$

$$\boxed{\frac{\partial \gamma(T_c)}{\partial U_j} = \gamma(T_c) \left[\frac{1}{T} \frac{\partial T}{\partial U_j} \left(\eta q + \frac{E_a q}{T_c} \right) \right]} \tag{7.32}$$

This works for any rate-controlling temperature, but for $T_c = T_{tr}$ (and hence $q = 1$), this reduces to:

$$\boxed{\frac{\partial \gamma(T)}{\partial U_j} = \gamma(T)\left[\frac{1}{T}\frac{\partial T}{\partial U_j}\left(\eta + \frac{E_a}{T}\right)\right]} \tag{7.33}$$

We also have the derivative of the equilibrium constant, $K_{c,m}$:

$$\frac{\partial K_{c,m}}{\partial U_j} = \frac{\partial K_{c,m}}{\partial T}\frac{\partial T}{\partial U_j} \tag{7.34}$$

$$\begin{aligned}\frac{\partial K_{c,m}}{\partial T} = \frac{\partial}{\partial T}\left[\left(\frac{p_0}{\hat{R}T}\right)^{\nu_m}\right]\exp\left[-\sum_{s=1}^{\text{NS}}(\nu_s'' - \nu_s')\left(\frac{H_s^\circ}{\hat{R}T} - \frac{S_s^\circ}{\hat{R}T}\right)\right] \\ + \left(\frac{p_0}{\hat{R}T}\right)^{\nu_m}\frac{\partial}{\partial T}\left\{\exp\left[-\sum_{s=1}^{\text{NS}}(\nu_s'' - \nu_s')\left(\frac{H_s^\circ}{\hat{R}T} - \frac{S_s^\circ}{\hat{R}T}\right)\right]\right\}\end{aligned} \tag{7.35}$$

We have:

$$\frac{\partial}{\partial T}\left[\left(\frac{p_0}{\hat{R}T}\right)^{\nu_m}\right] = \left(\frac{p_0}{\hat{R}T}\right)^{\nu_m}\left(-\frac{\nu_m}{T}\right) \tag{7.36}$$

$$\begin{aligned}&\frac{\partial}{\partial T}\left\{\exp\left[-\sum_{s=1}^{\text{NS}}(\nu_s'' - \nu_s')\left(\frac{H_s^\circ}{\hat{R}T} - \frac{S_s^\circ}{\hat{R}T}\right)\right]\right\} \\ &= \left\{\exp\left[-\sum_{s=1}^{\text{NS}}(\nu_s'' - \nu_s')\left(\frac{H_s^\circ}{\hat{R}T} - \frac{S_s^\circ}{\hat{R}T}\right)\right]\right\}\left\{\sum_{s=1}^{\text{NS}}(\nu_s'' - \nu_s')\frac{H_s^\circ}{\hat{R}T^2}\right\}\end{aligned} \tag{7.37}$$

Plugging these into the master equation gives:

$$\frac{\partial K_{c,m}(T)}{\partial U_j} = K_{c,m}(T)\left\{\frac{1}{T}\frac{\partial T}{\partial U_j}\left[\sum_s(\nu_s'' - \nu_s')\frac{H_s^\circ}{\hat{R}T} - \nu_m\right]\right\} \tag{7.38}$$

Since the equilibrium constant is formed with the backward rate temperature, and that is either T or T_e, no $\partial T_c/\partial T$ is needed. Finally, two derivatives based on the products over densities in the reaction rate equation:

$$\frac{\partial}{\partial \rho_j}\left[\frac{\rho_j}{\mathcal{M}_j}\right]^{\nu_j^*} = \nu_j^*\left[\frac{\rho_j}{\mathcal{M}_j}\right]^{\nu_j^*-1}\frac{1}{\mathcal{M}_j} \tag{7.39}$$

$$= \left[\frac{\rho_j}{\mathcal{M}_j}\right]^{\nu_j^*}\left[\frac{\nu_j^*}{\rho_j}\right] \tag{7.40}$$

and

$$\frac{\partial}{\partial \rho_j}\left\{\prod_{i=1}^{\text{NS}}\left[\frac{\rho_i}{\mathcal{M}_i}\right]^{\nu_i^*}\right\} = \nu_j^*\left[\frac{\rho_j}{\mathcal{M}_j}\right]^{\nu_j^*-1}\frac{1}{\mathcal{M}_j}\left\{\prod_{i\neq j}^{\text{NS}}\left[\frac{\rho_i}{\mathcal{M}_i}\right]^{\nu_i^*}\right\} \tag{7.41}$$

$$= \left\{\prod_{i=1}^{\text{NS}}\left[\frac{\rho_i}{\mathcal{M}_i}\right]^{\nu_i^*}\right\}\left[\frac{\nu_j^*}{\rho_j}\right] \tag{7.42}$$

Putting it together, the F term in the L_r derivative is:

$$F = \left\{\gamma(T_f)\left[\frac{1}{T}\frac{\partial T}{\partial U_j}\left(\eta q + \frac{E_a q}{T_f}\right)\right]\right\}\left[\frac{\rho_{s,f}}{\mathcal{M}_{s,f}}\right]^{\nu_s'} + \gamma(T_f)\left[\frac{\rho_{s,f}}{\mathcal{M}_{s,f}}\right]^{\nu_s'}\left[\frac{\nu_s^*}{\rho_s}\right] \tag{7.43}$$

$$\boxed{F = \gamma(T_f)\left[\frac{\rho_{s,f}}{\mathcal{M}_{s,f}}\right]^{\nu_s'}\left\{\left[\frac{1}{T}\frac{\partial T}{\partial U_j}\left(\eta q + \frac{E_a q}{T_f}\right)\right] + \underbrace{\frac{\nu_s'}{\rho_s}}_{\text{if } U_j = \rho_s}\right\}} \tag{7.44}$$

or:

$$\boxed{F = \gamma(T_f)\left[\frac{\rho_{s,f}}{\mathcal{M}_{s,f}}\right]^{\nu_s'}\left[\frac{1}{T}\frac{\partial T}{\partial U_j}\left(\eta q + \frac{E_a q}{T_f}\right)\right] + \underbrace{\gamma(T_f)\nu_s'\left[\frac{\rho_{s,f}}{\mathcal{M}_{s,f}}\right]^{\nu_s'-1}}_{\text{if } U_j = \rho_s}} \tag{7.45}$$

For the B term we must start by expanding the first term:

$$\left\{\frac{\partial}{\partial U_j}\left[\frac{\gamma(T_b)}{K_{c,m}(T_b)}\right]\right\}\left[\frac{\rho_{s,b}}{\mathcal{M}_{s,b}}\right]^{\nu_s''}$$

$$= \left\{\frac{\partial}{\partial U_j}\left[\gamma(T_b)\right]\frac{1}{K_{c,m}(T_b)} + \gamma(T_b)\frac{\partial}{\partial U_j}\left[\frac{1}{K_{c,m}(T_b)}\right]\right\}\left[\frac{\rho_{s,b}}{\mathcal{M}_{s,b}}\right]^{\nu_s''} \tag{7.46}$$

$$= \frac{\gamma(T_b)}{K_{c,m}(T_b)}\left[\frac{\rho_{s,b}}{\mathcal{M}_{s,b}}\right]^{\nu_s''}\left\{\frac{1}{T}\frac{\partial T}{\partial U_j}\left[\eta + \frac{E_a}{T} + \nu_m - \left(\sum_s(\nu_s'' - \nu_s')\frac{H_s^\circ}{\hat{R}T}\right)\right]\right\} \tag{7.47}$$

The second term in B is :

$$\frac{\gamma(T_b)}{K_c(T_b)}\left\{\frac{\partial}{\partial U_j}\left[\frac{\rho_{s,b}}{\mathcal{M}_{s,b}}\right]^{\nu_s''}\right\} = \frac{\gamma(T_b)}{K_c(T_b)}\left[\frac{\rho_{s,b}}{\mathcal{M}_{s,b}}\right]^{\nu_s''}\left[\frac{\nu_j''}{\rho_j}\right] \tag{7.48}$$

Like before, the only derivative that matters here is the one with respect to ρ_s. So this term is 0 unles U_j = ρ_s. Therefore, the final form of B is:

$$\boxed{B = \frac{\gamma(T_b)}{K_{c,m}(T_b)} \left[\frac{\rho_{s,b}}{\mathcal{M}_{s,b}}\right]^{\nu'_s} \left\{ \underbrace{\left[\frac{\nu''_j}{\rho_j}\right]}_{\text{if } U_j = \rho_s} + \frac{1}{T}\frac{\partial T}{\partial U_j}\left[\eta + \frac{E_a}{T} + \nu_m - \left(\sum_s (\nu''_s - \nu'_s)\frac{H_s^\circ}{\hat{R}T}\right)\right] \right\}} \tag{7.49}$$

or:

$$B = \frac{\gamma(T_b)}{K_{c,m}(T_b)} \prod_{i=1} \left[\frac{\rho_i}{\mathcal{M}_i}\right]^{\nu''_i} \left[\frac{1}{T}\frac{\partial T}{\partial U_j}\left[\eta + \frac{E_a}{T} + \nu_m - \left(\sum_s (\nu''_s - \nu'_s)\frac{H_s^\circ}{\hat{R}T}\right)\right]\right] + \frac{\gamma(T_b)}{K_{c,m}(T_b)}\nu'_s \left[\frac{\rho_{s,b}}{\mathcal{M}_{s,b}}\right]^{\nu''_s - 1} \tag{7.50}$$

$$\frac{\partial L_r}{\partial U_j} = \overbrace{\gamma(T_f)\left[\frac{\rho_{s,f}}{\mathcal{M}_{s,f}}\right]^{\nu'_s}}^{F} \left\{ \underbrace{\frac{\nu'_s}{\rho_s}}_{\text{if } U_j = \rho_s} + \frac{1}{T}\frac{\partial T}{\partial U_j}\left(\eta q + \frac{E_a q}{T_f}\right)\right\} - \underbrace{\frac{\gamma(T_b)}{K_{c,m}(T_b)}\left[\frac{\rho_{s,b}}{\mathcal{M}_{s,b}}\right]^{\nu'_s}}_{B} \left\{ \underbrace{\frac{\nu''_j}{\rho_j}}_{\text{if } U_j = \rho_s} + \frac{1}{T}\frac{\partial T}{\partial U_j}\left[\eta + \frac{E_a}{T} + \nu_m - \left(\sum_s (\nu''_s - \nu'_s)\frac{H_s^\circ}{\hat{R}T}\right)\right]\right\} \tag{7.51}$$

7.9.3 Source term Jacobians - Normal reactions

For normal reaction, we can revert to the usual notation of k_f and k_b for the forward and backward rate coefficients with no efficiencies. We also need the product across all concentrations. The following is the usual form:

$$R_r = \overbrace{\underbrace{k_f(T_f)\prod_i \left[\frac{\rho_i}{\mathcal{M}_i}\right]^{\nu'_i}}_{F} - \underbrace{\frac{k_f(T_b)}{K_c(T_b)}\prod_i \left[\frac{\rho_i}{\mathcal{M}_i}\right]^{\nu''_i}}_{B}}^{L_r} \tag{7.52}$$

The derivative of F and B wrt U_j is extremely similar to the third-body reaction ones:

$$\frac{\partial F}{\partial U_j} = k_f(T_F)\prod_{i=1}^{\text{NS}} \left[\frac{\rho_i}{\mathcal{M}_i}\right]^{\nu'_i} \left[\underbrace{\frac{\nu'_j}{\rho_j}}_{\text{if } U_j = \rho_i} + \frac{1}{T}\frac{\partial T}{\partial U_j}\left(\eta q + \frac{E_a q}{T_f}\right)\right] \tag{7.53}$$

$$\frac{\partial B}{\partial U_j} = \frac{k_f(T_b)}{K_c(T_b)} \prod_{i=1}^{\text{NS}} \left[\frac{\rho_i}{\mathcal{M}_i}\right]^{\nu''_i} \left[\underbrace{\frac{\nu''_j}{\rho_j}}_{\text{if } U_j = \rho_j} + \frac{1}{T}\frac{\partial T}{\partial U_j}\left(\eta + \frac{E_a}{T_b} + \nu_m - \sum s = 1^{\text{NS}}(\nu''_s - \nu'_s)\frac{H_s^\circ}{\hat{R}T}\right)\right] \tag{7.54}$$

The full derivative is:

$$\begin{aligned} \frac{\partial R_r}{\partial U_j} = k_f(T_F) \prod_{i=1}^{\text{NS}} \left[\frac{\rho_i}{\mathcal{M}_i}\right]^{\nu'_i} & \left[\underbrace{\frac{\nu'_j}{\rho_j}}_{\text{if } U_j = \rho_i} + \frac{1}{T}\frac{\partial T}{\partial U_j}\left(\eta q + \frac{E_a q}{T_f}\right)\right] \\ & - \frac{k_f(T_b)}{K_c(T_b)} \prod_{i=1}^{\text{NS}} \left[\frac{\rho_i}{\mathcal{M}_i}\right]^{\nu''_i} \left[\underbrace{\frac{\nu''_j}{\rho_j}}_{\text{if } U_j = \rho_j} + \frac{1}{T}\frac{\partial T}{\partial U_j}\left(\eta + \frac{E_a}{T_b} + \nu_m - \sum_{s=1}^{\text{NS}}(\nu''_s - \nu'_s)\frac{H_s^\circ}{\hat{R}T}\right)\right] \end{aligned} \tag{7.55}$$

7.9.4 Source term Jacobians - Vibrational relaxation

Recall the vibrational energy source term:

$$Q_{t-v,s} = \rho_s \frac{e_{vs}^* - e_{vs}}{\tau_s} \tag{7.56}$$

$$e_{vs} = \frac{\hat{R}}{\mathcal{M}_s} \frac{\theta_{vs}}{\exp\left(\theta_{vs}/T_v\right) - 1} \tag{7.57}$$

The derivative is then:

$$\begin{aligned} \frac{\partial Q_{t-v,s}}{\partial U_j} = \left[\frac{\partial}{\partial U_j}\rho_s\right] \frac{e_{vs}^* - e_{vs}}{\tau_s} & \\ + \rho_s \left\{ \left[\frac{\partial}{\partial U_j}(e_{vs}^* - e_{vs})\right]\frac{1}{\tau_s} + (e_{vs}^* - e_{vs})\left(-\tau_s^{-2}\left[\frac{\partial}{\partial U_j}\tau_s\right]\right)\right\} & \end{aligned} \tag{7.58}$$

Where we have the following:

$$\tau_s = < \tau_s > + \tau_s^P = \frac{\sum_{r=1}^{\text{NS}} X_r}{\sum_{r=1}^{\text{NS}} X_r/\tau_{sr}} + \tau_s^P \tag{7.59}$$

$$\frac{\partial e_{vs}}{\partial T_v} = \frac{\hat{R}}{\mathcal{M}_s}\left(\frac{\theta_{vs}}{T_v}\right)^2 \frac{\exp\left(\theta_{vs}/T_v\right)}{\left[\exp\left(\theta_{vs}/T_v\right) - 1\right]^2} \tag{7.60}$$

$$\frac{\partial e_{vs}^*}{\partial T_v} = \left.\frac{\partial e_{vs}}{\partial T_v}\right|_{T_v = T_{tr}} \tag{7.61}$$

We now aim to find the partial derivatives of the relaxation time:

$$\frac{\partial \tau_s}{\partial U_j} = \frac{\partial}{\partial U_j}\left[\frac{\sum_{r=1}^{\mathrm{NS}} X_r}{\sum_{r=1}^{\mathrm{NS}} X_r/\tau_{sr}} + \tau_s^P\right] \tag{7.62}$$

This gives:

$$\frac{\partial \tau_s}{\partial U_j} = \underbrace{\left[\frac{\partial}{\partial U_j}\sum_{r=1}^{\mathrm{NS}} X_r\right]\frac{1}{\sum_{r=1}^{\mathrm{NS}} X_r/\tau_{sr}}}_{0} + \sum_{r=1}^{\mathrm{NS}} X_r\left[\frac{\partial}{\partial U_j}\left(\sum_{r=1}^{\mathrm{NS}}\frac{X_r}{\tau_{sr}}\right)^{-1}\right] + \frac{\partial \tau_s^P}{\partial U_j} \tag{7.63}$$

Since the first term is zero, this leaves us with:

$$\frac{\partial \tau_s}{\partial U_j} = \left[\sum_{r=1}^{\mathrm{NS}} X_r\right]\left[\frac{\partial}{\partial U_j}\left(\sum_{r=1}^{\mathrm{NS}}\frac{X_r}{\tau_{sr}}\right)^{-1}\right] + \frac{\partial \tau_s^P}{\partial U_j} \tag{7.64}$$

Differentiating the second bracketed term:

$$\frac{\partial}{\partial U_j}\left(\sum_{r=1}^{\mathrm{NS}}\frac{X_r}{\tau_{sr}}\right)^{-1} = -\left(\sum_{r=1}^{\mathrm{NS}}\frac{X_r}{\tau_{sr}}\right)^{-2}\left[\frac{\partial}{\partial U_j}\sum_{r=1}^{\mathrm{NS}}\frac{X_r}{\tau_{sr}}\right] \tag{7.65}$$

Bringing the derivative inside, using the product rule and splitting the sums gives the final derivative as:

$$\frac{\partial \tau_s}{\partial U_j} = \frac{\partial \tau_s^P}{\partial U_j} - \left[\sum_{r=1}^{\mathrm{NS}} X_r\right]\left[\sum_{r=1}^{\mathrm{NS}}\frac{X_r}{\tau_{sr}}\right]^{-2} \times \ldots \left\{\sum_{r=1}^{\mathrm{NS}}\left[\frac{\partial X_r}{\partial U_j}\frac{1}{\tau_{sr}}\right] + \sum_{r=1}^{\mathrm{NS}}\left[X_r\left(\frac{\partial}{\partial U_j}\frac{1}{\tau_{sr}}\right)\right]\right\} \tag{7.66}$$

The blue term inside expands using an identity:

$$X_r = \frac{\rho_r/\mathcal{M}_r}{\sum_{i=1}^{\mathrm{NS}} \rho_i/\mathcal{M}_i} \tag{7.67}$$

$$\frac{\partial X_r}{\partial \rho_j} = \left[\frac{\partial}{\partial \rho_j}\frac{\rho_r}{\mathcal{M}_r}\right]\frac{1}{\sum_{i=1}^{\mathrm{NS}} \rho_i/\mathcal{M}_i} + \frac{\rho_r}{\mathcal{M}_r}\left[\frac{\partial}{\partial \rho_j}\frac{1}{\sum_{i=1}^{\mathrm{NS}} \rho_i/\mathcal{M}_i}\right] \tag{7.68}$$

Giving:

$$\frac{\partial X_r}{\partial \rho_j} = \frac{1}{\sum_{i=1}^{\mathrm{NS}} \rho_i/\mathcal{M}_i}\left[\frac{\delta_{jr}}{\mathcal{M}_r} - \frac{X_r}{\mathcal{M}_j}\right] \tag{7.69}$$

Noting that:

$$\frac{1}{\sum_{i=1}^{\text{NS}} \rho_i/\mathcal{M}_i} = \frac{\mathcal{M}_{\text{mix}}}{\rho_{\text{mix}}} \tag{7.70}$$

These mixture properties will be given no subscript going forward. This gives:

$$\boxed{\frac{\partial X_r}{\partial \rho_j} = \frac{\mathcal{M}}{\rho}\left[\frac{\delta_{jr}}{\mathcal{M}_r} - \frac{X_r}{\mathcal{M}_j}\right]} \tag{7.71}$$

And all other derivatives WRT conserved variables are 0. δ_{jr} is the Kronecker delta. Then $\rho_j = \rho_r$, it is 1, otherwise it is 0. We still need to sum this with the reciprocal of τ_{sr}. This produces:

$$\sum_{r=1}^{\text{NS}}\left[\frac{\partial X_r}{\partial U_j}\frac{1}{\tau_{sr}}\right] \tag{7.72}$$

$$\frac{\partial \tau_s}{\partial U_j} = \frac{\partial \tau_s^P}{\partial U_j} - \left[\sum_{r=1}^{\text{NS}} X_r\right]\left[\sum_{r=1}^{\text{NS}} \frac{X_r}{\tau_{sr}}\right]^{-2} \times \dots$$
$$\left\{\sum_{r=1}^{\text{NS}}\left[\frac{\partial X_r}{\partial U_j}\frac{1}{\tau_{sr}}\right] + \sum_{r=1}^{\text{NS}}\left[X_r\left(\frac{\partial}{\partial U_j}\frac{1}{\tau_{sr}}\right)\right]\right\} \tag{7.73}$$

The red term expands to the following:

$$\sum_{r=1}^{\text{NS}}\left[X_r\left(\frac{\partial}{\partial U_j}\frac{1}{\tau_{sr}}\right)\right] = -\sum_{r=1}^{\text{NS}}\left[X_r\tau_{sr}^{-2}\frac{\partial \tau_{sr}}{\partial U_j}\right] \tag{7.74}$$

The Millikan-White-only relaxation derivative is:

$$\boxed{\frac{\partial \tau_{sr}}{\partial U_j} = -\tau_{sr}\left[\frac{1}{p}\frac{\partial p}{\partial U_j} + \frac{1}{3}\frac{\partial T}{\partial U_j}T^{-4/3}A_{sr}\right]} \tag{7.75}$$

The derivatives of Park's high-temperature correction (in orange) can be found as:

$$\frac{\partial \tau_s^{\text{P}}}{\partial U_j} = \frac{\partial}{\partial U_j}\left[\frac{1}{\bar{c}_s\sigma n}\right] = -(\bar{c}_s\sigma n)^{-2}\left\{\frac{\partial \bar{c}_s}{\partial U_j}\sigma n + \bar{c}_s\left[\frac{\partial \sigma}{\partial U_j}n + \sigma\frac{\partial n}{\partial U_j}\right]\right\} \tag{7.76}$$

With:

$$\bar{c} = \sqrt{\frac{8R_sT}{\pi}}, \qquad n = \frac{p}{kT}, \qquad \sigma = \sigma_v'\left(\frac{50000}{T}\right)^2 \tag{7.77}$$

We begin with the $\bar{c}$ derivative:

$$\frac{\partial \bar{c}}{\partial U_j} = \sqrt{\frac{8R_s}{\pi}}\frac{\partial}{\partial U_j}\left[T^{1/2}\right] = \bar{c}_s\left[\frac{1}{2T}\frac{\partial T}{\partial U_j}\right] \tag{7.78}$$

The derivative of n is:

$$\frac{\partial n}{\partial U_j} = \frac{\partial p}{\partial U_j}\frac{1}{kT} + \frac{p}{k}\left(\frac{\partial}{\partial U_j}\frac{1}{T}\right) \tag{7.79}$$
$$= \frac{\partial p}{\partial U_j}\frac{1}{kT} - \frac{p}{k}\frac{1}{T^2}\frac{\partial T}{\partial U_j} \tag{7.80}$$

$$\boxed{\frac{\partial n}{\partial U_j} = n\left[\frac{1}{p}\frac{\partial p}{\partial U_j} - \frac{1}{T}\frac{\partial T}{\partial U_j}\right]} \tag{7.81}$$

Finally, the derivative wrt to σ:

$$\frac{\partial \sigma}{\partial U_j} = \frac{\partial}{\partial U_j}\left[\sigma_v'\left(\frac{50000}{T}\right)^2\right] \tag{7.82}$$
$$= \sigma_v' 50000^2 \frac{\partial}{\partial U_j}\frac{1}{T^2} \tag{7.83}$$
$$= \sigma_v' 50000^2 \left(-2T^{-3}\frac{\partial T}{\partial U_j}\right) \tag{7.84}$$

$$\boxed{\frac{\partial \sigma}{\partial U_j} = \sigma\left[-\frac{2}{T}\frac{\partial T}{\partial U_j}\right]} \tag{7.85}$$

Putting it all together for Park's derivative, we have:

$$\frac{\partial \tau_s^{\mathrm{P}}}{\partial U_j} = -(\bar{c}_s \sigma n)^{-2}\left\{\bar{c}_s\left[\frac{1}{2T}\frac{\partial T}{\partial U_j}\right]\sigma n \right.$$
$$\left. + \bar{c}\sigma\left[-\frac{2}{T}\frac{\partial T}{\partial U_j}\right]n + \bar{c}\sigma n\left[\frac{1}{p}\frac{\partial p}{\partial U_j} - \frac{1}{T}\frac{\partial T}{\partial U_j}\right]\right\} \tag{7.86}$$

Collecting terms, cancelling out, and flipping signs gives:

$$\boxed{\frac{\partial \tau_s^{\mathrm{P}}}{\partial U_j} = \tau_s^{\mathrm{P}}\left[\frac{5}{2T}\frac{\partial T}{\partial U_j} - \frac{1}{p}\frac{\partial p}{\partial U_j}\right]} \tag{7.87}$$

So, we have:

$$\frac{\partial \tau_s}{\partial U_j} = \tau_s^{\mathrm{P}}\left[\frac{5}{2T}\frac{\partial T}{\partial U_j} - \frac{1}{p}\frac{\partial p}{\partial U_j}\right] - \left[\sum_{r=1}^{\mathrm{NS}} X_r\right]\left[\sum_{r=1}^{\mathrm{NS}}\frac{X_r}{\tau_{sr}}\right]^{-2} \times \dots$$
$$\left\{\frac{\mathcal{M}}{\rho}\sum_{r=1}^{\mathrm{NS}}\left[\left(\frac{\delta_{jr}}{\mathcal{M}_r} - \frac{X_r}{\mathcal{M}_j}\right)\frac{1}{\tau_{sr}}\right] - \sum_{r=1}^{\mathrm{NS}}\left[X_r\tau_{sr}^{-2}\frac{\partial \tau_{sr}}{\partial U_j}\right]\right\} \tag{7.88}$$

$$\frac{\partial \tau_s}{\partial U_j} = \tau_s^{\mathrm{P}} \left[\frac{5}{2T} \frac{\partial T}{\partial U_j} - \frac{1}{p} \frac{\partial p}{\partial U_j} \right] - \left[\sum_{r=1}^{\mathrm{NS}} X_r \right] \left[\sum_{r=1}^{\mathrm{NS}} \frac{X_r}{\tau_{sr}} \right]^{-2} \times \dots$$
$$\left\{ \underbrace{\frac{\mathcal{M}}{\rho \mathcal{M}_j} \left[\frac{1}{\tau_{sj}} - \sum_{r=1}^{\mathrm{NS}} \frac{X_r}{\tau_{sr}} \right]}_{\text{when } U_j = \rho_j} - \sum_{r=1}^{\mathrm{NS}} \left[X_r \tau_{sr}^{-2} \frac{\partial \tau_{sr}}{\partial U_j} \right] \right\} \tag{7.89}$$

If we make some assumptions, such as:

$$\sum_{r=1}^{\mathrm{NS}} X_r = 1, \qquad < \tau_s > = \left[\sum_{r=1}^{\mathrm{NS}} \frac{X_r}{\tau_{sr}} \right]^{-1} \tag{7.90}$$

The above equation becomes:

$$\frac{\partial \tau_s}{\partial U_j} = \tau_s^{\mathrm{P}} \left[\frac{5}{2T} \frac{\partial T}{\partial U_j} - \frac{1}{p} \frac{\partial p}{\partial U_j} \right]$$
$$- < \tau_s >^2 \left\{ \underbrace{\frac{\mathcal{M}}{\rho \mathcal{M}_j} \left(\frac{1}{\tau_{sj}} - \frac{1}{< \tau_s >} \right)}_{\text{when } U_j = \rho_j} - \sum_{r=1}^{\mathrm{NS}} \left[\frac{X_r}{\tau_{sr}} \left(\frac{1}{p} \frac{\partial p}{\partial U_j} + \frac{1}{3} \frac{\partial T}{\partial U_j} T^{-4/3} A_{sr} \right) \right] \right\} \tag{7.91}$$

7.9.5 Explicit 0D Reactor

With no fluxes, the updated equation for the 0D reactor is:

$$\frac{U^{n+1} - U^n}{\Delta t} = S(U^n) \tag{7.92}$$

or:

$$\delta U = U^{n+1} - U^n, \qquad \delta U = \Delta t S(U^n) \tag{7.93}$$

7.9.6 Implicit 0D Reactor

The backward Euler time discretization for the 0D reactor look like:

$$\frac{U^{n+1} - U^n}{\Delta t} = S(U^{n+1}) \tag{7.94}$$

Linearize the source term as:

$$S(U^{n+1}) = S(U^n) + \frac{\partial S(U^n)}{\partial U_j} \delta U_j \tag{7.95}$$

And the final equation is:

$$[I - \Delta t J]\, \delta U = \Delta t S \tag{7.96}$$

Then:

$$U^{n+1} = U^n + \delta U \tag{7.97}$$

The nonlinear residual is:

$$R(U) = U - U^n - \Delta t S(U) \tag{7.98}$$

Where U = U^{n+1} or U^{k+1}. We aim to find a U that satisfies $R(U^{n+1})$ = 0. Therefore, we use Newton steps to find this value. First, take your current U_n, and set it as U_k.

8 Stagnation line solver

The system of equations for the stagnation line solver is:

$$\frac{\partial \boldsymbol{U}}{\partial t} + \frac{\partial \boldsymbol{F_c}}{\partial r} + \frac{\partial \boldsymbol{F_d}}{\partial r} = \boldsymbol{S_c} + \boldsymbol{S_d} + \boldsymbol{S_k} + \boldsymbol{S_r} \tag{8.1}$$

The conserved variable vector (**U**), primitive variable vector (**V**), convective flux vector ($\boldsymbol{F_c}$), and diffusive flux vector ($\boldsymbol{F_d}$) for the stagnation line solver are:

$$\boldsymbol{U} = \begin{bmatrix} \rho_s \\ \rho u_r \\ \rho u_\theta \\ \rho E \\ \rho e_v \end{bmatrix} \quad \boldsymbol{V} = \begin{bmatrix} \rho_s \\ u_r \\ u_\theta \\ p \\ e_v \end{bmatrix} \quad \boldsymbol{F_c} = \begin{bmatrix} \rho_s u_r \\ \rho u_r^2 + p \\ \rho u_r u_\theta \\ \rho u_r H \\ \rho u_r e_v \end{bmatrix} \quad \boldsymbol{F_d} = \begin{bmatrix} J_{rs} \\ -\tau_{rr} \\ -\tau_{r\theta} \\ q_r - \tau_{rr} u_r \\ q_r^{ve} \end{bmatrix} \tag{8.2}$$

The DRNSE also results in a convective source term ($\boldsymbol{S^c}$) and a diffusive source term ($\boldsymbol{S^d}$), along with the species-mass and vibrational-electronic energy source vector ($\boldsymbol{S^k}$), and the radiative source term ($\boldsymbol{S^r}$):

$$\boldsymbol{S_c} = -\frac{u_r + u_\theta}{r} \begin{bmatrix} 2\rho_s \\ 2\rho u_r \\ 3\rho u_\theta - 2\frac{p - p_\infty}{u_r + u_\theta} \\ 2\rho H \\ 2\rho e_{ve} \end{bmatrix} \quad \boldsymbol{S_d} = -\frac{2}{r} \begin{bmatrix} J_{rs} \\ \tau_{\theta\theta} - \tau_{rr} + \tau_{r\theta} \\ \frac{1}{2}(\tau_{\theta\theta} - 3\tau_{r\theta}) \\ q_r - \tau_{rr} u_r - \tau_{r\theta} u_r - \tau_{\theta\theta} u_\theta \\ q_r^{ve} \end{bmatrix} \tag{8.3}$$

$$\boldsymbol{S_k} = \begin{bmatrix} \dot{\omega}_s \\ 0 \\ 0 \\ 0 \\ \Omega_{ve} \end{bmatrix} \quad \boldsymbol{S_r} = \begin{bmatrix} \dot{\phi}_s \\ 0 \\ 0 \\ \mathcal{P} \\ \mathcal{P}_{ve} \end{bmatrix} \tag{8.4}$$

The terms in the diffusive flux and source vectors are as follows:

$$\begin{aligned}
\tau_{rr} &= \frac{4}{3}\mu \left(\frac{\partial u_r}{\partial r} - \frac{u_r + u_\theta}{r} \right) \\
\tau_{r\theta} &= \mu \left(\frac{\partial u_\theta}{\partial r} - \frac{u_r + u_\theta}{r} \right) \\
\tau_{\theta\theta} &= -\frac{1}{2}\tau_{rr} \\
q_r &= \sum_s J_{rs} h_s - \kappa^{tr} \frac{\partial T_{tr}}{\partial r} - \kappa^{ve} \frac{\partial T_{ve}}{\partial r} \\
q_r^{ve} &= \sum_s J_{rs} h_s^{ve} - \kappa^{ve} \frac{\partial T_{ve}}{\partial r}
\end{aligned}$$

The modified Fickian model for the species diffusion flux is used, which is a correction to the standard Fickian model to ensure that the sum of the species diffusion fluxes is zero (i.e., mass conservation is satisfied):

$$J_{rs} = -\rho D_s \frac{\partial Y_s}{\partial r} + Y_s \sum_{i=1}^{\mathrm{NS}} \left[\rho D_i \frac{\partial Y_i}{\partial r} \right] \tag{8.5}$$

8.1 Jacobians

In this section, we will derive all of the Jacobians that are necessary for the implicit time-stepping of the stagnation line solver. This includes the flux Jacobians, the source term Jacobians, and the Jacobian of the diffusive fluxes. There are quite a number of Jacobians, so a brief summary is given here:

- $\frac{\partial \boldsymbol{U}}{\partial \boldsymbol{V}}$ = Jacobian of conserved variables with respect to primitive variables.
- $\frac{\partial \boldsymbol{V}}{\partial \boldsymbol{U}}$ = Jacobian of primitive variables with respect to conserved variables.
- $\frac{\partial \boldsymbol{F_c}}{\partial \boldsymbol{V}}$ = Jacobian of convective flux vector with respect to primitive variables.
- $\tilde{\boldsymbol{R}}$ = Matrix of right eigenvectors of the convective flux Jacobian.
- $\tilde{\boldsymbol{L}}$ = Matrix of left eigenvectors of the convective flux Jacobian.
- $\boldsymbol{\Lambda}^{\pm}$ = Diagonal matrix of positive and negative eigenvalues of the convective flux Jacobian.
- $\boldsymbol{L}$ = Matrix of left eigenvectors of the convective flux Jacobian multiplied by the Jacobian of the primitive variables with respect to the conserved variables.
- $\boldsymbol{R}$ = Matrix of right eigenvectors of the convective flux Jacobian multiplied by the Jacobian of the conserved variables with respect to the primitive variables.
- $\boldsymbol{A}^{\pm}$ = Convective flux Jacobian $R\Lambda^{\pm}L$.
- $\boldsymbol{M}_D^F$ = Coefficient matrix that holds transport quantities for the diffusive flux vector.
- $\boldsymbol{B}_D^F$ = Coefficient vector that holds values not dependant on spatical gradients in the diffusive flux vector.
- $\boldsymbol{M}_D^S$ = Coefficient matrix that holds transport quantities for the diffusive source vector.
- $\boldsymbol{B}_D^S$ = Coefficient vector that holds values not dependant on spatical gradients in the diffusive source vector.
- $\boldsymbol{J}_D$ = Jacobian matrix $\partial \boldsymbol{W} / \partial \boldsymbol{U}$ use for mapping from viscous primitive vector to conserved variable vector.
- $\boldsymbol{J}_B^F$ = Jacobian of $\boldsymbol{B}_D^F$ with respect to $\boldsymbol{U}$.
- $\boldsymbol{J}_B^S$ = Jacobian of $\boldsymbol{B}_D^S$ with respect to $\boldsymbol{U}$.
- $\boldsymbol{J}_D^S$ = Jacobian of $\boldsymbol{S}_d$ with respect to $\boldsymbol{U}$.
- $\boldsymbol{J}_C^S$ = Jacobian of $\boldsymbol{S}_c$ with respect to $\boldsymbol{U}$.

8.1.1 Convective Flux Jacobians

In order to find the flux Jacobian $\boldsymbol{A} = \partial \boldsymbol{F_c}/\partial \boldsymbol{U}$, we need to compute a series of Jacobians. The first is the Jacobian of $\boldsymbol{U}$ with respect to $\boldsymbol{V}$. This is done by putting the conserved variables in terms of the primitive variables and then taking the derivatives. Then for the Jacobians of $\boldsymbol{V}$ with respect to $\boldsymbol{U}$, we do the same. By denoting ρ_s as either U_s or V_s, ρu_r as U_2, u_r as V_2, etc., we can write the vectors as:

$$\boldsymbol{V}(\boldsymbol{U}) = \begin{bmatrix} U_s \\ U_2/\sum_s U_s \\ U_3/\sum_s U_s \\ p(\boldsymbol{U}) \\ U_5/\sum_s U_s \end{bmatrix} \qquad \boldsymbol{U}(\boldsymbol{V}) = \begin{bmatrix} V_s \\ \sum_s V_s V_2 \\ \sum_s V_s V_3 \\ \rho E(\boldsymbol{V}) \\ \sum_s V_s V_5 \end{bmatrix} \qquad \boldsymbol{F_c}(\boldsymbol{V}) = \begin{bmatrix} V_s V_2 \\ \sum_s V_s V_2^2 + V_4 \\ \sum_s V_s V_2 V_3 \\ [\rho E(\boldsymbol{V}) + V_4] V_2 \\ \sum_s V_s V_2 V_5 \end{bmatrix}$$

Pressure is found by using the equation of state for a mixture of ideal gases and recovering translational energy from the total energy equation by subtracting out the other energy modes:

$$p = \rho \bar{R} T_{\text{tr}} = \frac{\bar{R}}{\bar{c}_{v,\text{tr}}} \left[\rho e^{\text{tr}}\right] = \frac{\bar{R}}{\bar{c}_{v,\text{tr}}} \left[\rho E - \frac{\rho}{2} u_r^2 - \rho e_v - \sum_s \rho_s h_s^f \right]$$

In conserved variable form, this is:

$$p(\boldsymbol{U}) = \frac{\sum U_s R_s}{\sum U_s c_{v,s}^{\text{tr}}} \left[U_4 - \frac{1}{2} \frac{1}{\sum_s U_s} U_2^2 - U_5 - \sum_s U_s h_s^{\circ} \right]$$

For internal energy, we can rearrange the above equation to get:

$$\rho E = \frac{\bar{c}_{v,\text{tr}}}{\bar{R}} p + \frac{1}{2} \rho u_r^2 + \rho e_v + \sum_s \rho_s h_s^f$$

In primitive variable form, this is:

$$\rho E(\boldsymbol{V}) = \frac{\sum_s V_s c_{v,s}^{\text{tr}}}{\sum_s V_s R_s} V_4 + \frac{1}{2} \sum_s V_s \cdot V_2^2 + \sum_s V_s \cdot V_5 + \sum_s V_s h_s^{\circ}$$

The derivates are now straightforward and the Jacobians are as follows:

$$\frac{\partial \boldsymbol{U}}{\partial \boldsymbol{V}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \ddots & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ u_r & \dots & u_r & \rho & 0 & 0 & 0 \\ u_\theta & \dots & u_\theta & 0 & \rho & 0 & 0 \\ \rho E_{\rho_1} & \dots & \rho E_{\rho_{\text{NS}}} & \rho E_{u_r} & \rho E_{u_\theta} & \rho E_p & \rho E_{e_v} \\ e_v & \dots & e_v & 0 & 0 & 0 & \rho \end{bmatrix}$$

$$\frac{\partial \boldsymbol{V}}{\partial \boldsymbol{U}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \ddots & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ -u_r/\rho & \dots & -u_r/\rho & 1/\rho & 0 & 0 & 0 \\ -u_\theta/\rho & \dots & -u_\theta/\rho & 0 & 1/\rho & 0 & 0 \\ p_{\rho_1} & \dots & p_{\rho_{\text{NS}}} & p_{\rho u_r} & p_{\rho u_\theta} & p_{\rho E} & p_{\rho e_v} \\ -e_v/\rho & \dots & -e_v/\rho & 0 & 0 & 0 & 1/\rho \end{bmatrix}$$

$$\frac{\partial \boldsymbol{F_c}}{\partial \boldsymbol{V}} = \begin{bmatrix} u_r & 0 & 0 & \rho_s & 0 & 0 & 0 \\ 0 & \ddots & 0 & \vdots & 0 & 0 & 0 \\ 0 & 0 & u_r & \rho_s & 0 & 0 & 0 \\ u_r^2 & \dots & u_r^2 & 2\rho u_r & 0 & 1 & 0 \\ u_r u_\theta & \dots & u_r u_\theta & \rho u_\theta & \rho u_r & 0 & 0 \\ \rho E_{\rho_1} u_r & \dots & \rho E_{\rho_{\text{NS}}} u_r & \rho E_{u_r} u_r + \rho H & \rho E_{u_\theta} u_r & (\rho E_p + 1) u_r & \rho E_{e_v} u_r \\ u_r e_v & \dots & u_r e_v & \rho e_v & 0 & 0 & \rho u_r \end{bmatrix}$$

Where we have the following derivatives for pressure and energy:

$$\begin{aligned} p_{\rho_s} &= \left[R_s - \frac{\bar{R}}{\bar{c}_{v,\text{tr}}} c_{v,s} \right] T_{\text{tr}} + \frac{\bar{R}}{\bar{c}_{v,\text{tr}}} \left[\frac{u_r^2}{2} - h_s^\circ \right] \\ p_{\rho u_r} &= -\frac{\bar{R}}{\bar{c}_{v,\text{tr}}} u_r \\ p_{\rho u_\theta} &= 0 \\ p_{\rho E} &= \frac{\bar{R}}{\bar{c}_{v,\text{tr}}} \\ p_{\rho e_v} &= -\frac{\bar{R}}{\bar{c}_{v,\text{tr}}} \end{aligned}$$

$$\begin{aligned} \rho E_{\rho_s} &= \left[c_{v,s} - \frac{\bar{c}_{v,\text{tr}}}{\bar{R}} R_s \right] T_{\text{tr}} + \frac{u_r^2}{2} + e_v + h_s^\circ \\ \rho E_{u_r} &= \rho u_r \\ \rho E_{u_\theta} &= 0 \\ \rho E_p &= \frac{\bar{c}_{v,\text{tr}}}{\bar{R}} \\ \rho E_{e_v} &= \rho \end{aligned}$$

The matrix product of $\partial \boldsymbol{V}/\partial \boldsymbol{U}$ and $\partial \boldsymbol{F_c}/\partial \boldsymbol{V}$ gives the flux Jacobian that needs to be diagonlized for the upwinding:

$$\frac{\partial \boldsymbol{V}}{\partial \boldsymbol{U}} \frac{\partial \boldsymbol{F_c}}{\partial \boldsymbol{V}} = \begin{bmatrix} u_r & 0 & 0 & \rho_s & 0 & 0 & 0 \\ 0 & \ddots & 0 & \vdots & 0 & 0 & 0 \\ 0 & 0 & u_r & \rho_s & 0 & 0 & 0 \\ 0 & \dots & 0 & u_r & 0 & 1/\rho & 0 \\ 0 & \dots & 0 & 0 & u_r & 0 & 0 \\ 0 & \dots & 0 & \rho a^2 & 0 & u_r & 0 \\ 0 & \dots & 0 & 0 & 0 & 0 & u_r \end{bmatrix}$$

Where ρa^2 is found as:

$$\rho a^2 = p \left(1 + \frac{\bar{R}}{\bar{c}_{v,\text{tr}}} \right)$$

The eigenvalues of this matrix are:

$$\lambda = \Big[u_r, \dots, u_r, u_r - a, u_r + a, u_r, u_r\Big]$$

The diagonal matrix of eigenvalues is:

$$\Lambda = \begin{bmatrix} \lambda_1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \ddots & 0 & 0 & 0 & 0 & 0 \\ 0 & \dots & \lambda_1 & 0 & 0 & 0 & 0 \\ 0 & \dots & 0 & \lambda_2 & 0 & 0 & 0 \\ 0 & \dots & 0 & 0 & \lambda_3 & 0 & 0 \\ 0 & \dots & 0 & 0 & 0 & \lambda_1 & 0 \\ 0 & \dots & 0 & 0 & 0 & 0 & \lambda_1 \end{bmatrix}$$

Where $\lambda_1 = u_r$, $\lambda_2 = u_r - a$, and $\lambda_3 = u_r + a$. The left and right matrix of eigenvectors are:

$$\tilde{\boldsymbol{L}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & -\frac{\rho_1}{\rho a^2} & 0 \\ 0 & \ddots & 0 & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 1 & 0 & 0 & -\frac{\rho_{N_s}}{\rho a^2} & 0 \\ 0 & \cdots & 0 & -\frac{1}{2a} & 0 & \frac{1}{2\rho a^2} & 0 \\ 0 & \cdots & 0 & \frac{1}{2a} & 0 & \frac{1}{2\rho a^2} & 0 \\ 0 & \cdots & 0 & 0 & 1 & 0 & 0 \\ 0 & \cdots & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \qquad \tilde{\boldsymbol{R}} = \begin{bmatrix} 1 & 0 & 0 & \rho_1 & \rho_1 & 0 & 0 \\ 0 & \ddots & 0 & \vdots & \vdots & 0 & 0 \\ 0 & \dots & 1 & \rho_{\text{NS}} & \rho_{\text{NS}} & 0 & 0 \\ 0 & \dots & 0 & -a & a & 0 & 0 \\ 0 & \dots & 0 & 0 & 0 & 1 & 0 \\ 0 & \dots & 0 & \rho a^2 & \rho a^2 & 0 & 0 \\ 0 & \dots & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

These matrices are subsequently multiplied by the primitive/consvered variable Jacobians as $\boldsymbol{L} = \tilde{L}\frac{\partial V}{\partial U}$ and $\boldsymbol{R} = \frac{\partial U}{\partial V}\tilde{R}$ to get the final left and right eigenvector matrices that are used for the upwinding of the fluxes. These matrices are:

$$\boldsymbol{L} = \tilde{\boldsymbol{L}}\frac{\partial \boldsymbol{V}}{\partial \boldsymbol{U}} = \begin{bmatrix} 1 - \frac{Y_1 p_{\rho_1}}{a^2} & -\frac{Y_1 p_{\rho_s}}{a^2} & -\frac{Y_1 p_{\rho_{\text{NS}}}}{a^2} & \frac{Y_s \bar{R} u_r}{a^2 \bar{c}_v} & 0 & -\frac{Y_s \bar{R}}{a^2 \bar{c}_v} & \frac{Y_s \bar{R}}{a^2 \bar{c}_v} \\ -\frac{Y_s p_{\rho_1}}{a^2} & \ddots & -\frac{Y_s p_{\rho_{\text{NS}}}}{a^2} & \frac{Y_s \bar{R} u_r}{a^2 \bar{c}_v} & 0 & -\frac{Y_s \bar{R}}{a^2 \bar{c}_v} & \frac{Y_s \bar{R}}{a^2 \bar{c}_v} \\ -\frac{Y_{\text{NS}} p_{\rho_1}}{a^2} & -\frac{Y_{\text{NS}} p_{\rho_s}}{a^2} & 1 - \frac{Y_{\text{NS}} p_{\rho_{\text{NS}}}}{a^2} & \frac{Y_s \bar{R} u_r}{a^2 \bar{c}_v} & 0 & -\frac{Y_s \bar{R}}{a^2 \bar{c}_v} & \frac{Y_s \bar{R}}{a^2 \bar{c}_v} \\ \frac{1}{2\rho a}\left[\frac{p_{\rho_1}}{a} + u_r\right] & \dots & \frac{1}{2\rho a}\left[\frac{p_{\rho_{\text{NS}}}}{a} + u_r\right] & -\frac{1}{2\rho a}\left[1 + \frac{\bar{R} u_r}{\bar{c}_v a}\right] & 0 & \frac{\bar{R}}{2\rho a^2 \bar{c}_v} & -\frac{\bar{R}}{2\rho a^2 \bar{c}_v} \\ \frac{1}{2\rho a}\left[\frac{p_{\rho_1}}{a} - u_r\right] & \dots & \frac{1}{2\rho a}\left[\frac{p_{\rho_{\text{NS}}}}{a} - u_r\right] & \frac{1}{2\rho a}\left[1 - \frac{\bar{R} u_r}{\bar{c}_v a}\right] & 0 & \frac{\bar{R}}{2\rho a^2 \bar{c}_v} & -\frac{\bar{R}}{2\rho a^2 \bar{c}_v} \\ -\frac{u_\theta}{\rho} & \dots & -\frac{u_\theta}{\rho} & 0 & \frac{1}{\rho} & 0 & 0 \\ -\frac{e_v}{\rho} & \dots & -\frac{e_v}{\rho} & 0 & 0 & 0 & \frac{1}{\rho} \end{bmatrix}$$

$$\boldsymbol{R} = \frac{\partial \boldsymbol{U}}{\partial \boldsymbol{V}}\tilde{\boldsymbol{R}} = \begin{bmatrix} 1 & 0 & 0 & \rho_1 & \rho_1 & 0 & 0 \\ 0 & \ddots & 0 & \vdots & \vdots & 0 & 0 \\ 0 & \dots & 1 & \rho_{\text{NS}} & \rho_{\text{NS}} & 0 & 0 \\ u_r & \dots & u_r & \rho(u_r - a) & \rho(u_r + a) & 0 & 0 \\ u_\theta & \dots & u_\theta & \rho u_\theta & \rho u_\theta & \rho & 0 \\ \rho E_{\rho_1} & \dots & \rho E_{\rho_{\text{NS}}} & \rho(H - u_r a) & \rho(H + u_r a) & 0 & \rho \\ e_v & \dots & e_v & \rho e_v & \rho e_v & 0 & \rho \end{bmatrix}$$

8.1.2 Diffusive Flux Jacobians

The explicit diffusive flux computed by using a coefficient matrix, a Jacobian matrix, a vector of derivatives, and a vector of fluxes not dependant on gradients:

$$\boldsymbol{F}_d = \boldsymbol{M}_D^F \boldsymbol{J}_D \frac{\partial \boldsymbol{U}}{\partial r} + \boldsymbol{B}_D^F$$

For implicit time integration, a Jacobian of the $\boldsymbol{B}_D^F$ term is also needed to linearize the flux as:

$$\boldsymbol{F}_d^{n+1} = \boldsymbol{F}_d^n + \delta \boldsymbol{F}_d = \boldsymbol{M}_D^F \boldsymbol{J}_D \frac{\partial \boldsymbol{U}^n}{\partial r} + \boldsymbol{B}_D^F + \boldsymbol{M}_D^F \boldsymbol{J}_D \frac{\partial \delta \boldsymbol{U}}{\partial r} + \boldsymbol{J}_B \delta \boldsymbol{U}$$

We will go through each matrix here. The coefficient matrix $\boldsymbol{M}$ contains the viscous coefficients:

$$\boldsymbol{M}_D^F = \begin{bmatrix} D_1(\rho_1 - \rho) & \rho_1 D_s & \rho_1 D_{\text{NS}} & 0 & 0 & 0 & 0 \\ \rho_s D_1 & D_s(\rho_s - \rho) & \rho_s D_{\text{NS}} & 0 & 0 & 0 & 0 \\ \rho_{\text{NS}} D_1 & \rho_{\text{NS}} D_s & D_{\text{NS}}(\rho_{\text{NS}} - \rho) & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{4}{3}\mu & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\mu & 0 & 0 \\ a^1 & b^1 & c^1 & -\frac{4}{3}\mu u_r & 0 & -\kappa^{tr} & -\kappa^{ve} \\ a^{ve} & b^{ve} & c^{ve} & 0 & 0 & 0 & -\kappa^{ve} \end{bmatrix}$$

Where we have the following definitions for a, b, and c:

$$\begin{aligned} a^m &= h_1^m D_1(\rho_1 - \rho) + \sum_{k \neq 1} h_k^m \rho_k D_1 \\ b^m &= h_s^m D_s(\rho_s - \rho) + \sum_{k \neq s} h_k^m \rho_k D_s \\ c^m &= h_{\text{NS}}^m D_{\text{NS}}(\rho_{\text{NS}} - \rho) + \sum_{k \neq \text{NS}} h_k^m \rho_k D_{\text{NS}} \end{aligned}$$

Superscript m denotes which enthalpy is being used. For a^1, we use the total specific enthalpy. For a^{ve}, we use the vibrational-electronic specific enthalpy defined as $h_s^{ve} = e_s^{ve}$. The same applies for b and c. These equations are essentially just the contributions from each species towards the diffusive fluxes multiplied by the species enthalpy using the modified Fickian model for mass diffusion. The Jacobians $\boldsymbol{J}_D$ maps primitive variable gradients to conserved variable gradients. The following primitive variable vector is used:

$$\boldsymbol{W} = \begin{bmatrix} Y_s & u_r & u_\theta & T_{tr} & T_{ve} \end{bmatrix}^T$$

Therefore, the Jacobian $\boldsymbol{J}_D$ is defined as $\frac{\partial \boldsymbol{W}}{\partial \boldsymbol{U}}$. As before, we must write the primitive variables in terms of the conserved variables to take the derivatives:

$$\boldsymbol{W}(\boldsymbol{U}) = \begin{bmatrix} U_s/\sum_s U_s \\ U_2/\sum_s U_s \\ U_3/\sum_s U_s \\ T_{tr}(\boldsymbol{U}) \\ T_{ve}(\boldsymbol{U}) \end{bmatrix}$$

The translational termperature is found by rearranging the equation for total energy and subtracting out the contributions from the other energy modes:

$$T_{tr} = \left[\rho E - \rho e_{ve} - \frac{\rho}{2}u_r^2 - \sum_s \rho_s h_s^\circ\right] \frac{1}{\sum_s \rho_s c_{v,s}^{\text{tr}}}$$

In conserved variable form, this is:

$$T_{tr}(\boldsymbol{U}) = \left[U_4 - U_5 - \frac{1}{2}\frac{U_2^2}{\sum_s U_s} - \sum_s U_s h_s^\circ\right] \frac{1}{\sum_s U_s c_{v,s}^{\text{tr}}}$$

It is then straight corward to compute the derivative for T_{tr}. The derivatives are shown below:

$$\begin{aligned}
\frac{\partial T_{tr}}{\partial \rho_j} &= \frac{1}{\rho \bar{c}_{v,\text{tr}}}\left[\frac{1}{2}u_r^2 - h_j^\circ - c_{v,j}^{\text{tr}} T_{tr}\right] \\
\frac{\partial T_{tr}}{\partial \rho u_r} &= -\frac{u_r}{\rho \bar{c}_{v,\text{tr}}} \\
\frac{\partial T_{tr}}{\partial \rho u_\theta} &= 0 \\
\frac{\partial T_{tr}}{\partial \rho E} &= \frac{1}{\rho \bar{c}_{v,\text{tr}}} \\
\frac{\partial T_{tr}}{\partial \rho e_v} &= -\frac{1}{\rho \bar{c}_{v,\text{tr}}}
\end{aligned}$$

For the vibrational-electronic temperature, we use implicit differentiation to find the derivative. We start with the equation for vibrational-electronic energy:

$$\rho e_{ve} = \sum_s \rho_s e_s^{ve}$$

By subtracting the right hand side from the left hand side, we can set the equation equal to zero:

$$\rho e_{ve} - \sum_s \rho_s e_s^{ve} = 0$$

By taking the derivative with respect to U_i, we can find the Jacobian entries for the vibrational-electronic temperature implicitly:

$$\frac{\partial \rho e_{ve}}{\partial U_i} - \sum_s \frac{\partial \rho_s}{\partial U_i} e_s^{ve} - \sum_s \rho_s \frac{\partial e_s^{ve}}{\partial T_{ve}} \frac{\partial T_{ve}}{\partial U_i} = 0 \tag{8.6}$$

Rearranging this equation to solve for $\frac{\partial T_{ve}}{\partial U_i}$ gives:

$$\frac{\partial T_{ve}}{\partial U_i} = \frac{\frac{\partial \rho e_{ve}}{\partial U_i} - \sum_s \frac{\partial \rho_s}{\partial U_i} e_s^{ve}}{\sum_s \rho_s \frac{\partial e_s^{ve}}{\partial T_{ve}}}$$

This gives the following derivatives of T_{ve} with respect to the conserved variables, where $\rho c_v^{ve} = \sum_s \rho_s \frac{\partial e_s^{ve}}{\partial T_{ve}}$ is the mass weighted vibrational-electronic specific heat:

$$\frac{\partial T_{ve}}{\partial \rho_j} = -\frac{e_j^{ve}}{\rho c_v^{ve}}, \qquad \frac{\partial T_{ve}}{\partial \rho e_v} = \frac{1}{\rho c_v^{ve}} \tag{8.7}$$

$$\frac{\partial T_{ve}}{\partial \rho u_r} = \frac{\partial T_{ve}}{\partial \rho u_\theta} = \frac{\partial T_{ve}}{\partial \rho E} = 0 \tag{8.8}$$

With these results for the temperature derivatives, we can write the full Jacobian $\boldsymbol{J}_D$ as:

$$\boldsymbol{J}_D = \frac{\partial \boldsymbol{W}}{\partial \boldsymbol{U}} = \begin{bmatrix} \frac{1}{\rho}[1-Y_1] & -\frac{Y_1}{\rho} & -\frac{Y_1}{\rho} & 0 & 0 & 0 & 0 \\ -\frac{Y_s}{\rho} & \frac{1}{\rho}[1-Y_s] & -\frac{Y_s}{\rho} & 0 & 0 & 0 & 0 \\ -\frac{Y_{\mathrm{NS}}}{\rho} & -\frac{Y_{\mathrm{NS}}}{\rho} & \frac{1}{\rho}[1-Y_{\mathrm{NS}}] & 0 & 0 & 0 & 0 \\ -\frac{u_r}{\rho} & -\frac{u_r}{\rho} & -\frac{u_r}{\rho} & \frac{1}{\rho} & 0 & 0 & 0 \\ -\frac{u_\theta}{\rho} & -\frac{u_\theta}{\rho} & -\frac{u_\theta}{\rho} & 0 & \frac{1}{\rho} & 0 & 0 \\ \frac{\partial T_{tr}}{\partial \rho_s} & \dots & \frac{\partial T_{tr}}{\partial \rho_s} & \frac{\partial T_{tr}}{\partial \rho u_r} & \frac{\partial T_{tr}}{\partial \rho u_\theta} & \frac{\partial T_{tr}}{\partial \rho E} & \frac{\partial T_{tr}}{\partial \rho e_v} \\ \frac{\partial T_{ve}}{\partial \rho_s} & \dots & \frac{\partial T_{ve}}{\partial \rho_s} & 0 & 0 & 0 & \frac{\partial T_{ve}}{\partial \rho e_v} \end{bmatrix}$$

The matrix $\boldsymbol{B}_D^F$ that is not dependant on the spatial gradients is:

$$\boldsymbol{B}_D^F = \frac{1}{r}\begin{bmatrix} 0 & 0 & 0 & \frac{4}{3}\mu\hat{u} & \mu\hat{u} & \frac{4}{3}\mu u_r \hat{u} & 0 \end{bmatrix}^T$$

Again, we must put this in terms of the conserved variables to take the derivatives for the Jacobian $\boldsymbol{J}_B^F$:

$$\boldsymbol{B}_D^F = \frac{1}{r}\begin{bmatrix} 0 & 0 & 0 & \frac{4}{3}\mu\frac{U_2+U_3}{\sum_s U_s} & \mu\frac{U_2+U_3}{\sum_s U_s} & \frac{4}{3}\mu\frac{U_2^2+U_2U_3}{(\sum_s U_s)^2} & 0 \end{bmatrix}^T$$

The derivatives are straightforward to compute and the Jacobian $\boldsymbol{J}_B^F$ is:

$$\boldsymbol{J}_B^F = \frac{\partial \boldsymbol{B}}{\partial \boldsymbol{U}} = \frac{1}{r}\frac{\mu}{\rho}\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{4}{3}\hat{u} & -\frac{4}{3}\hat{u} & -\frac{4}{3}\hat{u} & \frac{4}{3} & \frac{4}{3} & 0 & 0 \\ -\hat{u} & -\hat{u} & -\hat{u} & 1 & 1 & 0 & 0 \\ -\frac{8}{3}u_r\hat{u} & -\frac{8}{3}u_r\hat{u} & -\frac{8}{3}u_r\hat{u} & \frac{4}{3}(\hat{u}+u_r) & \frac{4}{3}u_r & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{8.9}$$

Where $\hat{u} = u_r + u_\theta$.

8.1.3 Source Jacobians

We must first take source terms and write them in terms of the conserved variables:

$$\boldsymbol{S_c} = \begin{bmatrix} -2U_s \frac{U_2+U_3}{\sum_s U_s r} \\ -2U_2 \frac{U_2+U_3}{\sum_s U_s r} \\ 3U_3 \frac{U_2+U_3}{\sum_s U_s r} - 2\frac{p(\boldsymbol{U})-p_\infty}{r} \\ -2(U_4 + p(\boldsymbol{U}))\frac{U_2+U_3}{\sum_s U_s r} \\ -2U_5 \frac{U_2+U_3}{\sum_s U_s r} \end{bmatrix} \qquad \boldsymbol{S_d} = -\frac{2}{r}\begin{bmatrix} J_{rs} \\ -2\mu\frac{\partial u_r}{\partial r} + \mu\frac{\partial u_\theta}{\partial r} \\ -\frac{1}{3}\mu\frac{\partial u_r}{\partial r} - \frac{3}{2}\mu\frac{\partial u_\theta}{\partial r} \\ q_r + \left[\frac{2}{3}\mu u_\theta - \frac{4}{3}\mu u_r\right]\frac{\partial u_r}{\partial r} - \mu u_r \frac{\partial u_\theta}{\partial r} \\ q_r^{ve} \end{bmatrix} + \frac{\mu}{r}\hat{u}\begin{bmatrix} 0 \\ 1 \\ \frac{11}{6} \\ \frac{7}{3}u_r - \frac{2}{3}u_\theta \\ 0 \\ 0 \end{bmatrix}$$

First, the convective source term Jacobian is computed as:

$$\boldsymbol{J}_C^S = \frac{\partial \boldsymbol{S_c}}{\partial \boldsymbol{U}} = -\frac{2}{r}\begin{bmatrix} (\delta_{sj} - Y_s)\hat{u} & Y_s & Y_s & 0 & 0 \\ -u_r\hat{u} & u_r + \hat{u} & u_r & 0 & 0 \\ -\frac{3}{2}u_\theta\hat{u} - p_{\rho_s} & \frac{3}{2}u_\theta + \frac{\bar{R}}{\bar{c}_{v,\mathrm{tr}}}u_r & \frac{3}{2}(u_\theta + \hat{u}) & -\frac{\bar{R}}{\bar{c}_{v,\mathrm{tr}}} & \frac{\bar{R}}{\bar{c}_{v,\mathrm{tr}}} \\ \hat{u}(p_{\rho_s} - H) & H - \frac{\bar{R}}{\bar{c}_{v,\mathrm{tr}}}u_r\hat{u} & H & \hat{u}(1 + \frac{\bar{R}}{\bar{c}_{v,\mathrm{tr}}}) & -\frac{\bar{R}}{\bar{c}_{v,\mathrm{tr}}}\hat{u} \\ -\hat{u}e_{ve} & e_{ve} & e_{ve} & 0 & \hat{u} \end{bmatrix}$$

The diffusive source term is again split up as $\boldsymbol{S}_d = -\frac{2}{r}\left[\boldsymbol{M}_D^S \boldsymbol{J}_D \frac{\partial \boldsymbol{U}}{\partial r} + \boldsymbol{B}_D^S\right]$, giving the following matrices and vectors:

$$\boldsymbol{M}_D^S = \begin{bmatrix} D_1(\rho_1 - \rho) & \rho_1 D_s & \rho_1 D_{\mathrm{NS}} & 0 & 0 & 0 & 0 \\ \rho_s D_1 & D_s(\rho_s - \rho) & \rho_s D_{\mathrm{NS}} & 0 & 0 & 0 & 0 \\ \rho_{\mathrm{NS}} D_1 & \rho_{\mathrm{NS}} D_s & D_{\mathrm{NS}}(\rho_{\mathrm{NS}} - \rho) & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2\mu & \mu & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{3}\mu & -\frac{3}{2}\mu & 0 & 0 \\ a^1 & b^1 & c^1 & \frac{2}{3}\mu(u_\theta - 2u_r) & -\mu u_r & -\kappa^{tr} & -\kappa^{ve} \\ a^{ve} & b^{ve} & c^{ve} & 0 & 0 & 0 & -\kappa^{ve} \end{bmatrix}$$

$$\boldsymbol{B}_D^S = -\frac{\mu}{r}\hat{u}\begin{bmatrix} 0 & 0 & 0 & 1 & \frac{11}{6} & \frac{1}{3}(7u_r + 2u_\theta) & 0 & 0 \end{bmatrix}^T$$

$$\boldsymbol{J}_B^S = \frac{\mu}{\rho r}\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hat{u} & \hat{u} & \hat{u} & -1, & -1 & 0 & 0 \\ \frac{11}{6}\hat{u} & \frac{11}{6}\hat{u} & \frac{11}{6}\hat{u} & -\frac{11}{6} & -\frac{11}{6} & 0 & 0 \\ \frac{2}{3}\hat{u}(7u_r + 2u_\theta) & \frac{2}{3}\hat{u}(7u_r + 2u_\theta) & \frac{2}{3}\hat{u}(7u_r + 2u_\theta) & -\frac{1}{3}(14u_r + 9u_\theta) & -\frac{1}{3}(9u_r + 4u_\theta) & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

8.2 Solving the system of equations

8.2.1 Explicit time stepping

8.2.2 Implicit time stepping

The solution procedure for the implicit time stepping uses linearization of the flux and source terms. The linearized terms are found by using a truncated Taylor series expansion:

$$\boldsymbol{F}^{n+1}_{c,i+\frac{1}{2}} = \underbrace{\boldsymbol{A}^{+}_{i+\frac{1}{2}}\boldsymbol{U}_i + \boldsymbol{A}^{-}_{i+\frac{1}{2}}\boldsymbol{U}_{i+1}}_{\boldsymbol{F}_{c,i+\frac{1}{2}}} + \underbrace{\boldsymbol{A}^{+}_{i+\frac{1}{2}}\delta\boldsymbol{U}^{n+1}_i + \boldsymbol{A}^{-}_{i+\frac{1}{2}}\delta\boldsymbol{U}^{n+1}_{i+1}}_{\delta\boldsymbol{F}_{c,i+\frac{1}{2}}}$$

$$\boldsymbol{S_c}^{n+1} = \boldsymbol{S}_{c,i} + \boldsymbol{J}^S_{c,i}\delta\boldsymbol{U}^{n+1}_i \tag{8.10}$$

$$\begin{aligned}\boldsymbol{F}^{n+1}_{d,i+\frac{1}{2}} = {} & \overbrace{\boldsymbol{M}^F_{D,i+\frac{1}{2}}\boldsymbol{J}_{D,i+\frac{1}{2}}\frac{\boldsymbol{U}_{i+1}-\boldsymbol{U}_i}{\Delta r_i} + \boldsymbol{B}^F_{D,i+\frac{1}{2}}}^{\boldsymbol{F}_{d,i+\frac{1}{2}}} \\ & + \underbrace{\boldsymbol{M}^F_{D,i+\frac{1}{2}}\boldsymbol{J}_{D,i+\frac{1}{2}}\frac{\delta\boldsymbol{U}^{n+1}_{i+1}-\delta\boldsymbol{U}^{n+1}_i}{\Delta r_i} + \frac{1}{2}\left[\boldsymbol{J}^F_{B,i}\delta\boldsymbol{U}^{n+1}_i + \boldsymbol{J}^F_{B,i+1}\delta\boldsymbol{U}^{n+1}_{i+1}\right]}_{\delta\boldsymbol{F}_{d,i+\frac{1}{2}}}\end{aligned}$$

$$\boldsymbol{S}^{n+1}_{d,i} = \underbrace{\boldsymbol{M}^S_{D,i+\frac{1}{2}}\boldsymbol{J}_{D,i+\frac{1}{2}}\frac{\boldsymbol{U}_{i+1}-\boldsymbol{U}_{i-1}}{2\Delta r_i} + \boldsymbol{B}^S_{D,i}}_{\boldsymbol{S}_{d,i}} + \underbrace{\boldsymbol{M}^S_{D,i+\frac{1}{2}}\boldsymbol{J}_{D,i+\frac{1}{2}}\frac{\delta\boldsymbol{U}^{n+1}_{i+1}-\delta\boldsymbol{U}^{n+1}_{i-1}}{2\Delta r_i} + \boldsymbol{J}^S_{B,i}\delta\boldsymbol{U}^{n+1}_i}_{\delta\boldsymbol{S}_{d,i}}$$

$$\boldsymbol{S}^{n+1}_{k,i} = \boldsymbol{S}_{k,i} + \frac{\partial\boldsymbol{S}_{k,i}}{\partial\boldsymbol{U}_i}\delta\boldsymbol{U}^{n+1}_i$$

The updated equation for the implicit time stepping is:

$$\frac{\delta\boldsymbol{U}^{n+1}_i}{\Delta t}\Delta r_i = \boldsymbol{F}^n_{c,i-\frac{1}{2}} + \boldsymbol{A}^{+}_{i-\frac{1}{2}}\delta\boldsymbol{U}^{n+1}_{i-1} + \boldsymbol{A}^{-}_{i-\frac{1}{2}}\delta\boldsymbol{U}^{n+1}_i \tag{8.11}$$

$$+ \boldsymbol{F}^n_{d,i-\frac{1}{2}} + \frac{1}{\Delta r_i}\left(\boldsymbol{M}\boldsymbol{J}^F_d\delta\boldsymbol{U}^{n+1}_i - \boldsymbol{M}\boldsymbol{J}^F_d\delta U^{n+1}_{i-1}\right) \tag{8.12}$$

$$- \left[\boldsymbol{F}^n_{c,i+\frac{1}{2}} + \boldsymbol{A}^{+}_{i+\frac{1}{2}}\delta\boldsymbol{U}^{n+1}_i + \boldsymbol{A}^{-}_{i+\frac{1}{2}}\delta\boldsymbol{U}^{n+1}_{i+1}\right] \tag{8.13}$$

$$- \left[\boldsymbol{F}^n_{d,i-\frac{1}{2}} + \frac{1}{\Delta r_i}\left(\boldsymbol{M}\boldsymbol{J}^F_d\delta\boldsymbol{U}^{n+1}_i - \boldsymbol{M}\boldsymbol{J}^F_d\delta U^{n+1}_{i-1}\right)\right] \tag{8.14}$$

$$+ \boldsymbol{S}^n_{c,i} + \boldsymbol{J}^S_c\delta\boldsymbol{U}^{n+1}_i + \boldsymbol{S}^n_{d,i} + \boldsymbol{J}^S_d\delta\boldsymbol{U}^{n+1}_i + \boldsymbol{S}^n_{k,i} + \boldsymbol{J}^S_k\delta\boldsymbol{U}^{n+1}_i + \boldsymbol{S}^n_{r,i} + \boldsymbol{J}^S_r\delta\boldsymbol{U}^{n+1}_i \tag{8.15}$$

Solving this equation for the update δU gives:

$$\boldsymbol{A}\delta\boldsymbol{U}^{n+1}_{i+1} + \boldsymbol{B}\delta\boldsymbol{U}^{n+1}_i + \boldsymbol{C}\delta\boldsymbol{U}^{n+1}_{i-1} = \boldsymbol{D} \tag{8.16}$$

Where we have the following definitions for the matrices $\boldsymbol{A}$, $\boldsymbol{B}$, $\boldsymbol{C}$, and $\boldsymbol{D}$:

$$\boldsymbol{A} = \underbrace{\boldsymbol{A}^{-}_{i+\frac{1}{2}}}_{\text{Convective}} + \underbrace{\frac{1}{\Delta r_i} \boldsymbol{M} J^F_d \big|_{i+\frac{1}{2}}}_{\text{Diffusive}} \tag{8.17}$$

$$\boldsymbol{B} = \frac{\Delta r_i}{\Delta t} \boldsymbol{I} + \underbrace{\boldsymbol{A}^{+}_{i+\frac{1}{2}} - \boldsymbol{A}^{-}_{i-\frac{1}{2}} - \Delta r_i \boldsymbol{J}^S_c}_{\text{Convective}} \underbrace{- \frac{1}{\Delta r_i} \left(\boldsymbol{M} J^F_d \big|_{i-\frac{1}{2}} + \boldsymbol{M} J^F_d \big|_{i+\frac{1}{2}} \right) - \Delta r_i \boldsymbol{J}^S_d}_{\text{Diffusive}} \tag{8.18}$$

$$\boldsymbol{C} = \frac{1}{\Delta r_i} \boldsymbol{M} J^F_d \big|_{i-\frac{1}{2}} - \boldsymbol{A}^{+}_{i-\frac{1}{2}} \tag{8.19}$$

$$\boldsymbol{D} = \boldsymbol{F}^n_{c,i-\frac{1}{2}} - \boldsymbol{F}^n_{c,i+\frac{1}{2}} + \boldsymbol{F}^n_{d,i-\frac{1}{2}} - \boldsymbol{F}^n_{d,i+\frac{1}{2}} + \Delta r_i \left(\boldsymbol{S}^n_{c,i} + \boldsymbol{S}^n_{d,i} + \boldsymbol{S}^n_{k,i} + \boldsymbol{S}^n_{r,i} \right) \tag{8.20}$$

Which gives a block tridiagonal system of equations to be solved for:

8.3 Boundary Conditions

Boundary conditions are needed for both the explicit and implicit time stepping. For explicit time stepping, the boundary conditions are applied using ghost cells. For implicit time stepping, the boundary conditions are applied by modifying the update $\delta \boldsymbol{U}$ for the inner cell at the boundaries. We will go through each of these cases for both inviscid and viscous boundary conditions.

Each explicit boundary conditions ensures that properties are guaranteed to be satisfied at the boundary face. For an inflow, the ghost cell is set such that the primitive variables at the boundary face are equal to the specified primitive variables. For an outflow, the ghost cell is set such that the primitive variables at the boundary face are equal to the primitive variables at the inner cell such that there are no gradients at the outflow. For a symmetry boundary, the ghost cell is set such that the normal velocity at the boundary face is 0 and the other primitive variables at the boundary face are equal to the primitive variables at the inner cell.

For implicit boundary conditions, the update $\delta \boldsymbol{U}$ for the ghost cell is folded back into the update for the inner cell. For inflows, the update $\delta \boldsymbol{U}$ for the inner cell is set to 0 since the primitive variables are specified. For outflows, the update $\delta \boldsymbol{U}$ for the inner cell is modified such that there are no gradients at the boundary face. For symmetry boundaries, the update $\delta \boldsymbol{U}$ for the inner cell is modified such that the normal velocity at the boundary face is 0 and there are no gradients for the other primitive variables at the boundary face. In all cases, the ghost cell update is either all 0s, or the ghost cell update is some linear combination of the inner cell update, therefore the inner cell update can be modified to account for the ghost cell update as:

$$\delta \boldsymbol{U}_i = [\boldsymbol{I} \pm \boldsymbol{E}]\, \delta \boldsymbol{U}_i \tag{8.21}$$

First-order perturbations are used to generate the $\boldsymbol{E}$ matrix for each boundary condition. These derivations will be shown below.

8.3.1 Inviscid Explicit Boundary Conditions

8.3.2 Viscous Explicit Boundary Conditions

8.3.3 Inviscid Implicit Boundary Conditions

For inflows, the primitive variables are specified, therefore the update $\delta\boldsymbol{U}$ is 0, and hence the $\boldsymbol{E}$ matrix is also 0 since it has no dependance on the inner cell's update.

For symmetry boundaries, the normal velocity at the boundary face is 0. Densities and energies remain the same, as well as u_θ, but the u_r velocity component is flipped across the boundary. These are the only two inviscid matrices that are needed since there are no outflow boundary conditions in a stagnation-line solver. The $\boldsymbol{E}$ matrices are shown below.

$$\boldsymbol{E}^C_{\text{symmetry}} = \begin{bmatrix} 1 & \dots & 0 & 0 & 0 & 0 & 0 \\ \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \dots & 1 & 0 & 0 & 0 & 0 \\ 0 & \dots & 0 & -1 & 0 & 0 & 0 \\ 0 & \dots & 0 & 0 & 1 & 0 & 0 \\ 0 & \dots & 0 & 0 & 0 & 1 & 0 \\ 0 & \dots & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\boldsymbol{E}^C_{\text{inflow}} = \begin{bmatrix} 0 & \dots & 0 & 0 & 0 & 0 & 0 \\ \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \dots & 0 & 0 & 0 & 0 & 0 \\ 0 & \dots & 0 & 0 & 0 & 0 & 0 \\ 0 & \dots & 0 & 0 & 0 & 0 & 0 \\ 0 & \dots & 0 & 0 & 0 & 0 & 0 \\ 0 & \dots & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

8.3.4 Viscous Implicit Boundary Conditions

For an inflow, we retain the same matrix of 0s as before. Likewise for a symmetry boundary condition. However, for a no-slip isothermal or adibatic wall, the primitive variables at the face or ghost cell are specified. For temperature, we have:

$$T_g = 2T_w - T_i$$

By perturbing this equation, we can find the entry for $\boldsymbol{E}$ for the temperature update:

$$\begin{aligned} \delta T_g &= 2\delta T_w - \delta T_i \\ \delta T_g &= -\delta T_i \end{aligned}$$

The pressure in the ghost cell is set equal to the inner cell, therefore we must update the density in the ghost cell to satisfy the equation of state. Perturbing the equation of state gives:

$$
\begin{aligned}
\rho_g T_g &= \rho_i T_i \\
\rho_g &= \rho_i \frac{T_i}{2T_w - T_i} \\
\rho_g &= \rho_i f \\
\delta\rho_g &= \delta\rho_i f + \rho_i \frac{\partial f}{\partial T_i} \delta T_i \\
\delta\rho_g &= \left[\frac{T_i}{2T_w - T_i}\right] \delta\rho_i + \left[\rho_i \frac{1}{2T_w - T_i}\left(1 + \frac{T_i}{2T_w - T_i}\right)\right] \delta T_i
\end{aligned}
$$

Since the velocity at the wall is 0, the momentum in the ghost cell is set equal to the negative of the momentum in the inner cell. Perturbing this gives:

$$
\begin{aligned}
\delta(\rho u)_g &= -\delta(\rho u)_i \\
\delta(\rho u)_g &= -\delta(\rho u)_i
\end{aligned}
$$

The final $\boldsymbol{E}$ matrix for a no-slip isothermal wall is shown below:

$$
\boldsymbol{E}^V_{\text{isothermal}} = \begin{bmatrix}
T_i/T_g & \dots & 0 & 0 & 0 & \alpha & 0 \\
\vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \vdots \\
0 & \dots & T_i/T_g & 0 & 0 & \alpha & 0 \\
0 & \dots & 0 & -1 & 0 & 0 & 0 \\
0 & \dots & 0 & 0 & -1 & 0 & 0 \\
0 & \dots & 0 & 0 & 0 & -1 & 0 \\
0 & \dots & 0 & 0 & 0 & 0 & -1
\end{bmatrix}
$$

Where:

$$
\alpha = \frac{\rho_{s,i}}{T_g}\left(1 + \frac{T_i}{T_g}\right), \qquad T_g = 2T_w - T_i \tag{8.22}
$$

For an adiabatic wall, the temperature in the ghost cell is set equal to the temperature in the inner cell. Because of this, the density in the ghost cell is also set equal to the density in the inner cell to satisfy the equation of state. No-slip is still enfored so the final matrix is:

$$
\boldsymbol{E}^V_{\text{isothermal}} = \begin{bmatrix}
1 & \dots & 0 & 0 & 0 & 0 & 0 \\
\vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \vdots \\
0 & \dots & 1 & 0 & 0 & 0 & 0 \\
0 & \dots & 0 & -1 & 0 & 0 & 0 \\
0 & \dots & 0 & 0 & -1 & 0 & 0 \\
0 & \dots & 0 & 0 & 0 & 1 & 0 \\
0 & \dots & 0 & 0 & 0 & 0 & 1
\end{bmatrix}
$$